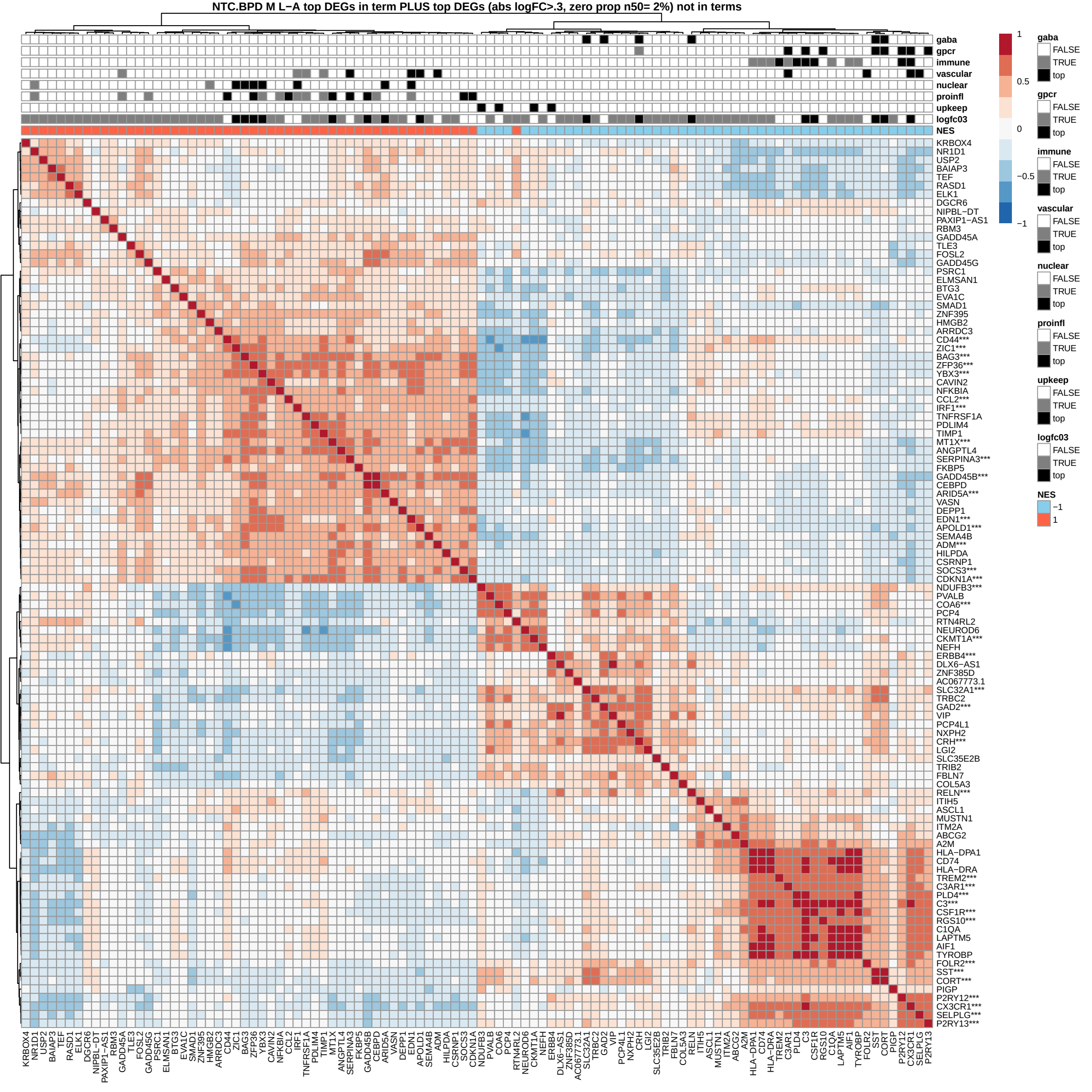
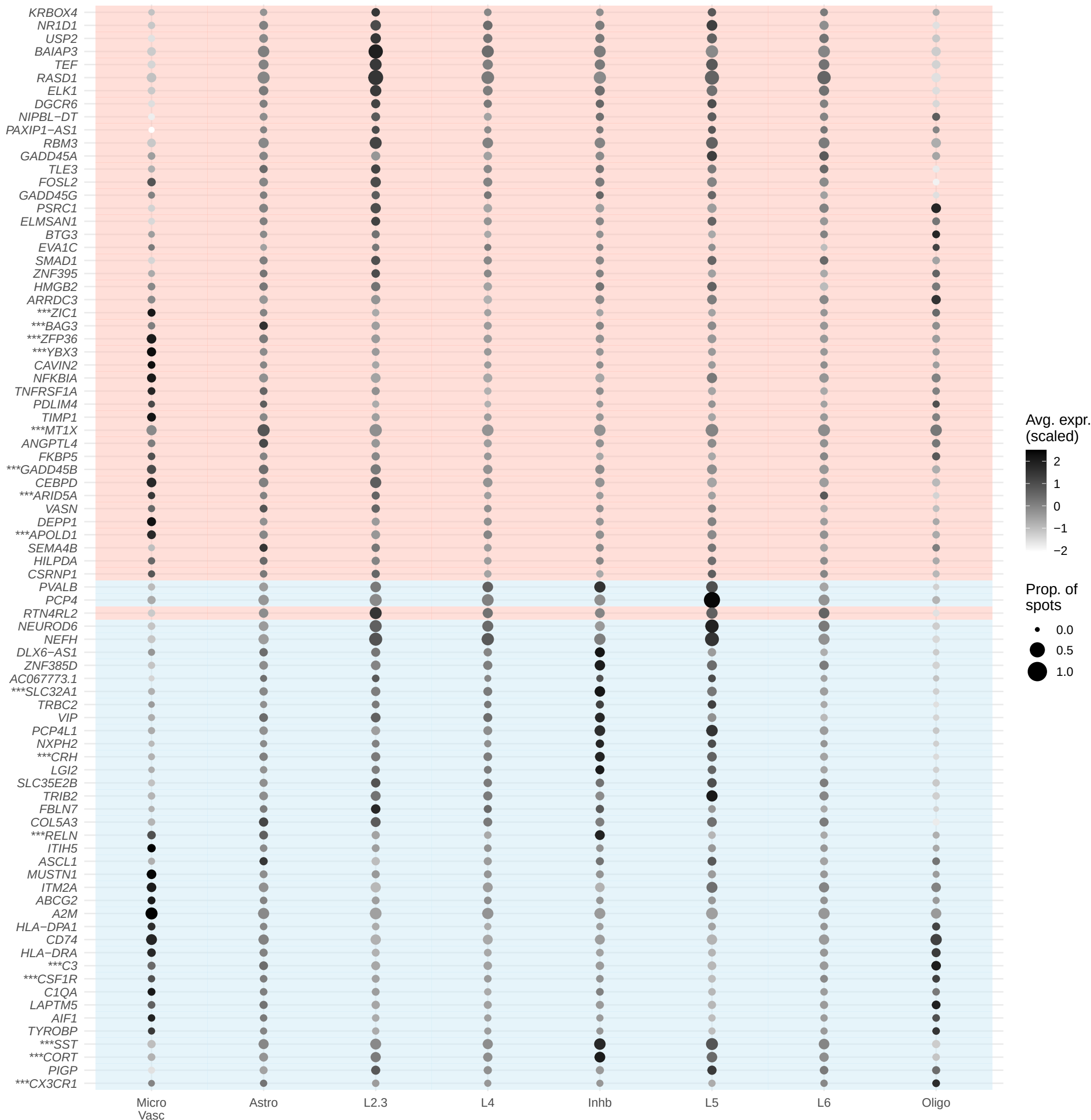


NTC.BPD M L-A top DEGs in term PLUS top DEGs (abs logFC>.3, zero prop n50= 2%) not in terms



NTC.BPD M L-A DEGs with abs(logFC)>.3 and zero prop.>2% in half of capture areas
(background indicates whether term was enriched/depleted)

GSEA term genes

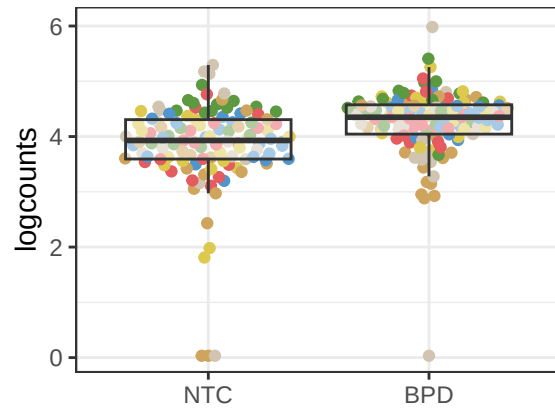


All genes with $\text{abs}(\log\text{FC}) > .5$

KRBOX4

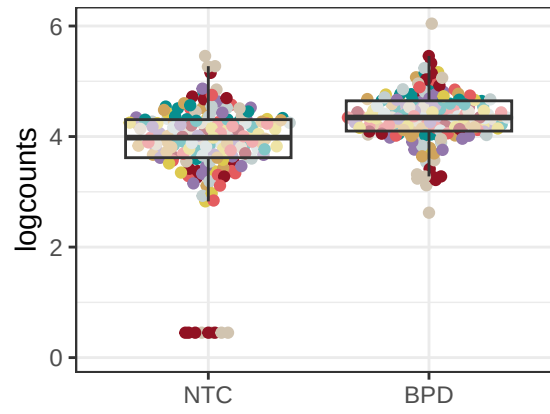
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}03$ logFC = 0.41



Seurat labels

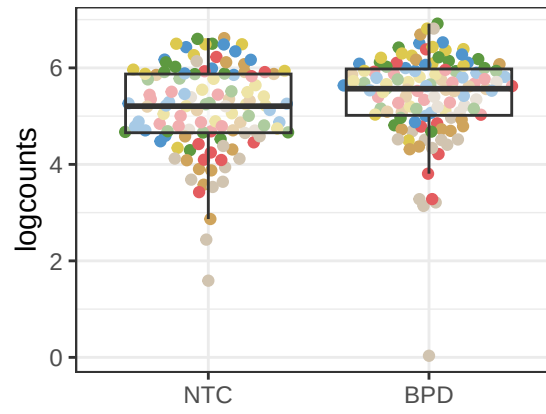
L-A sig: adj p < $3\text{e-}03$ logFC = 0.39



NR1D1

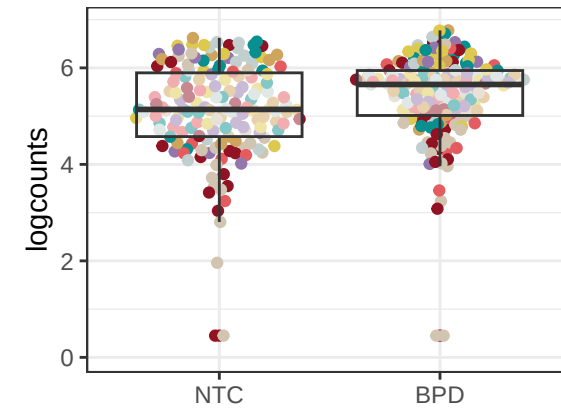
PRECAST (smoothed)

L-A sig: adj p < $1\text{e-}02$ logFC = 0.39



Seurat labels

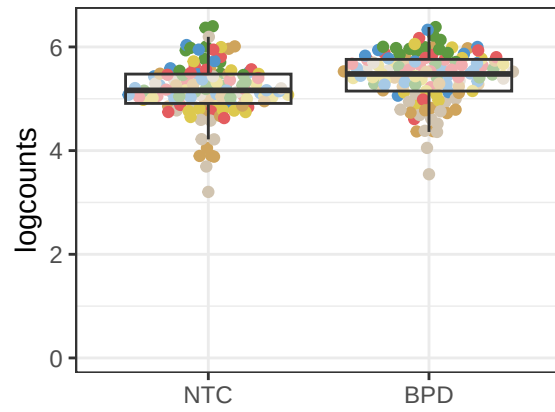
L-A sig: adj p < $1\text{e-}02$ logFC = 0.38



USP2

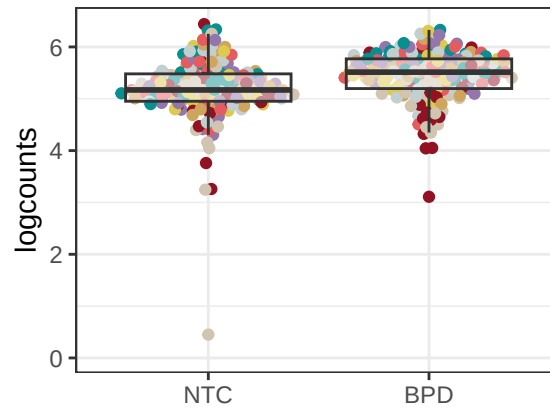
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}03$ logFC = 0.33



Seurat labels

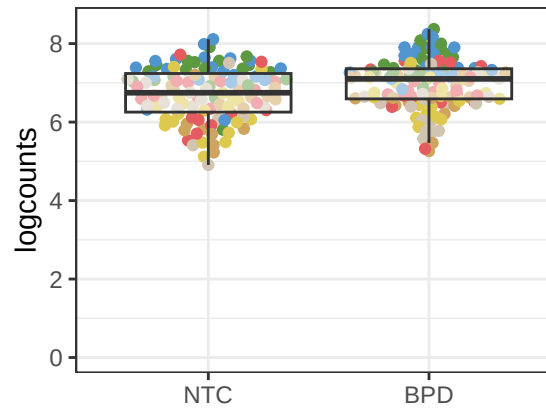
L-A sig: adj p < $7\text{e-}04$ logFC = 0.35



BAIAP3

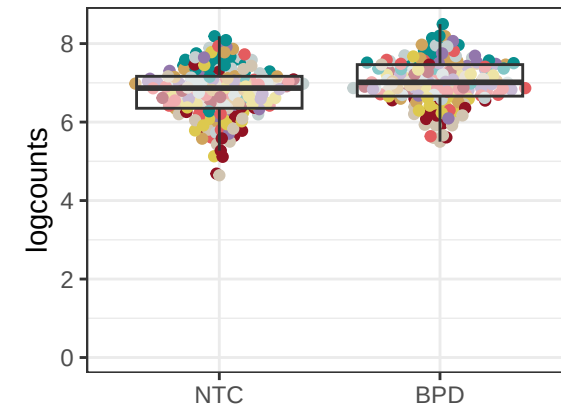
PRECAST (smoothed)

L-A sig: adj p < $1\text{e-}03$ logFC = 0.33



Seurat labels

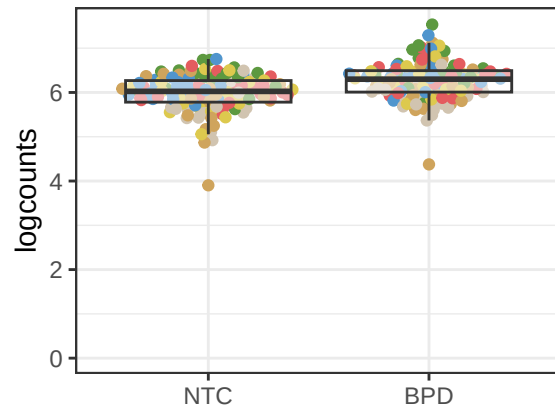
L-A sig: adj p < $9\text{e-}04$ logFC = 0.3



TEF

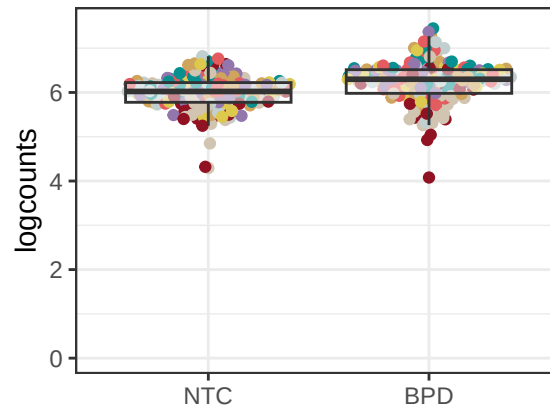
PRECAST (smoothed)

L-A sig: adj p < $4\text{e-}05$ logFC = 0.32



Seurat labels

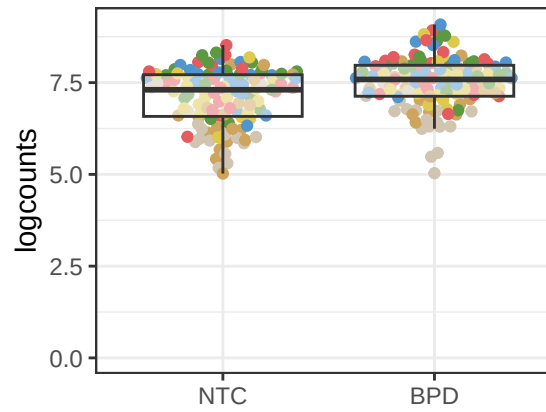
L-A sig: adj p < $3\text{e-}05$ logFC = 0.34



RASD1

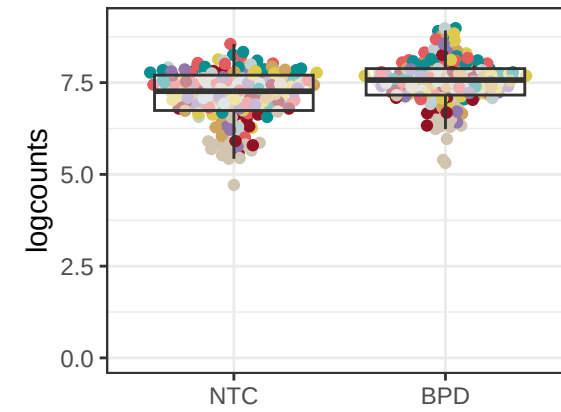
PRECAST (smoothed)

L-A sig: adj p < $4\text{e-}03$ logFC = 0.38



Seurat labels

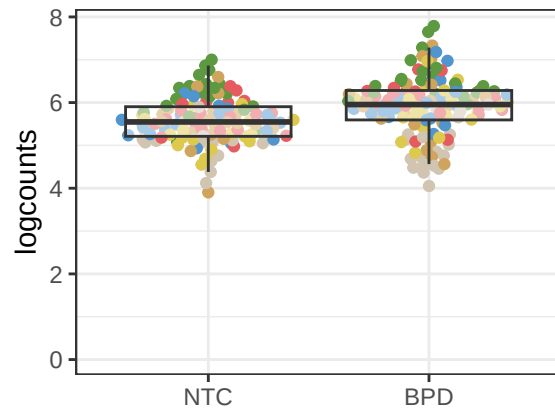
L-A sig: adj p < $2\text{e-}04$ logFC = 0.39



ELK1

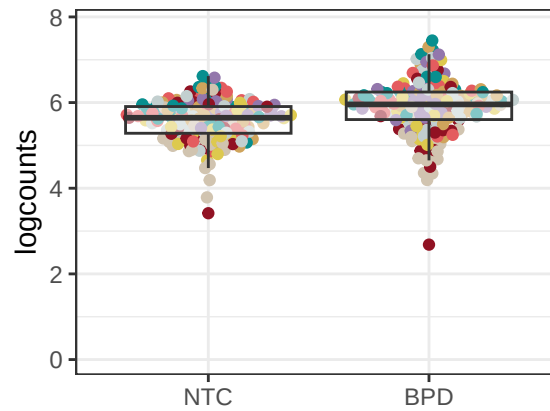
PRECAST (smoothed)

L-A sig: adj p < $5\text{e-}06$ logFC = 0.47



Seurat labels

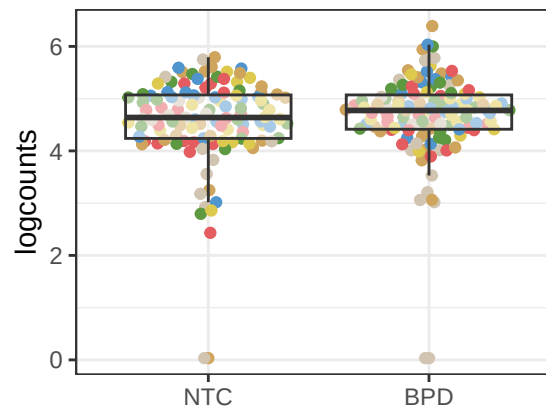
L-A sig: adj p < $4\text{e-}07$ logFC = 0.48



DGCR6

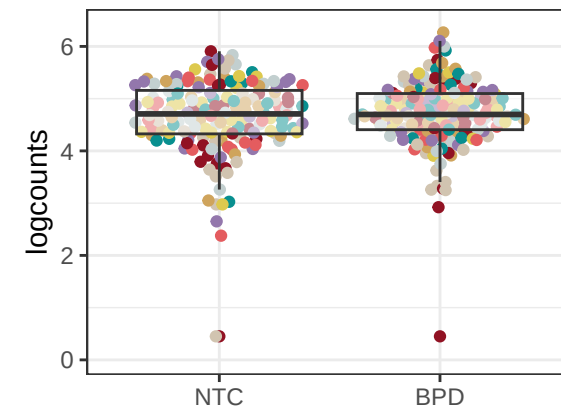
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}02$ logFC = 0.3



Seurat labels

L-A sig: adj p < $2\text{e-}02$ logFC = 0.32

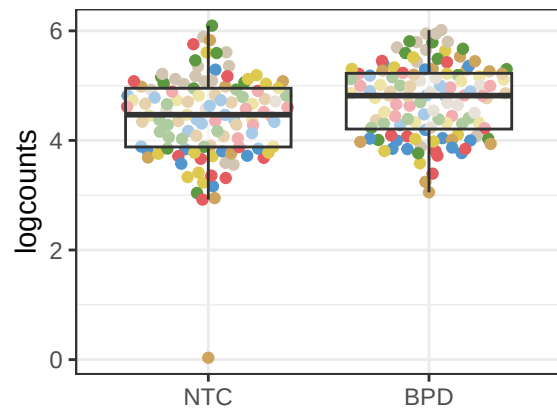


All genes with $\text{abs}(\log\text{FC}) > .5$

NIPBL-DT

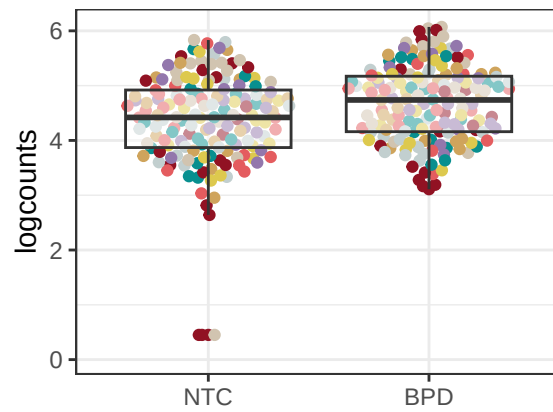
PRECAST (smoothed)

L-A sig: adj p < $9\text{e-}03$ logFC = 0.42



Seurat labels

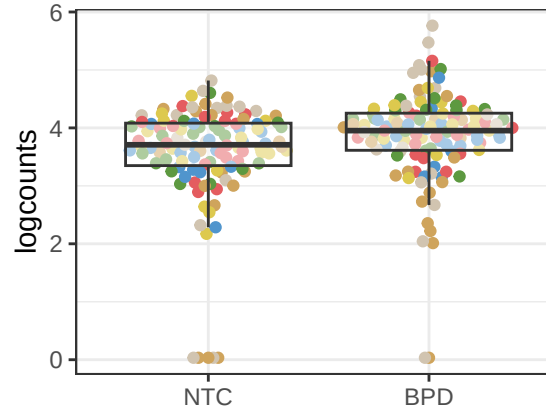
L-A sig: adj p < $1\text{e-}02$ logFC = 0.42



PAXIP1-AS1

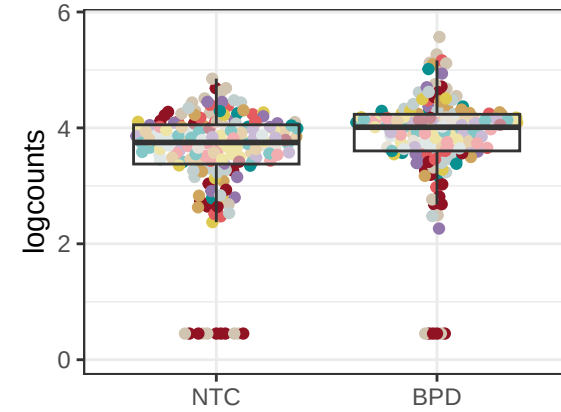
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}02$ logFC = 0.36



Seurat labels

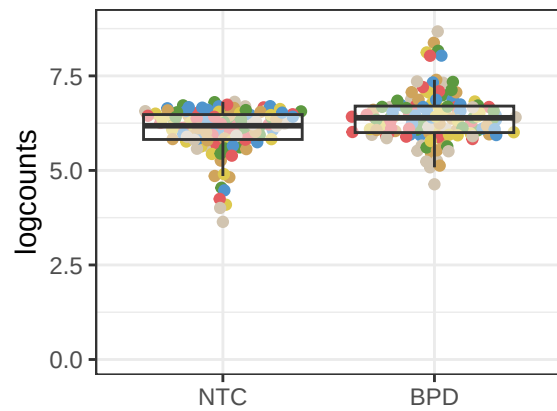
L-A sig: adj p < $5\text{e-}02$ logFC = 0.36



RBM3

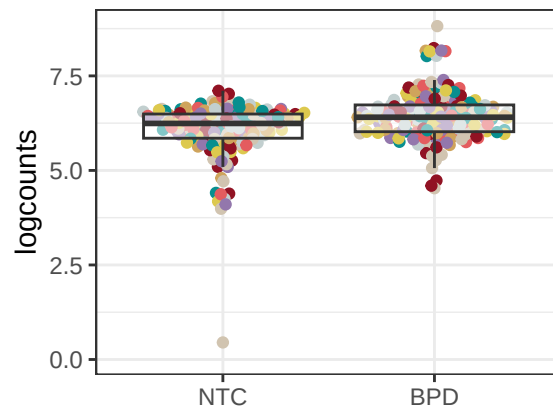
PRECAST (smoothed)

L-A sig: adj p < $6\text{e-}05$ logFC = 0.4



Seurat labels

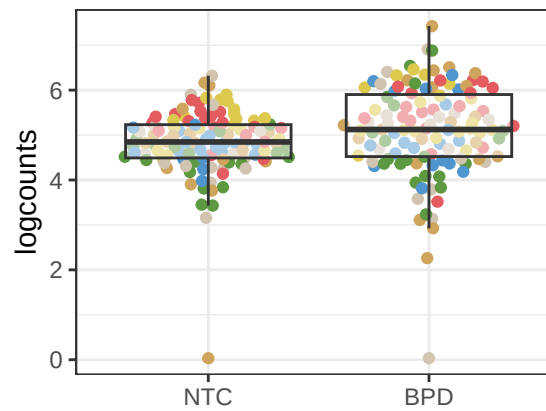
L-A sig: adj p < $3\text{e-}05$ logFC = 0.38



GADD45A

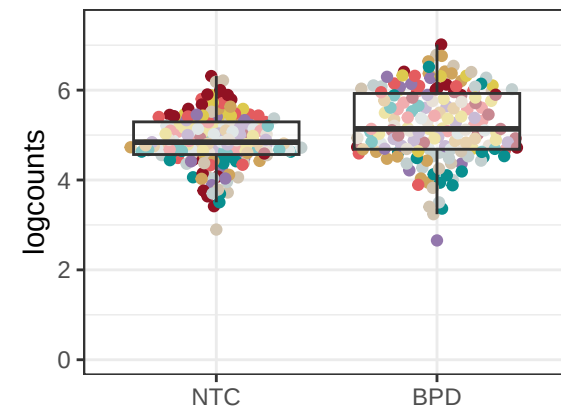
PRECAST (smoothed)

L-A sig: adj p < $4\text{e-}02$ logFC = 0.42



Seurat labels

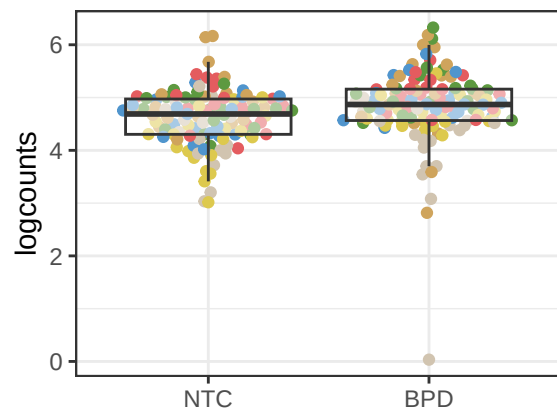
L-A sig: adj p < $2\text{e-}02$ logFC = 0.44



TLE3

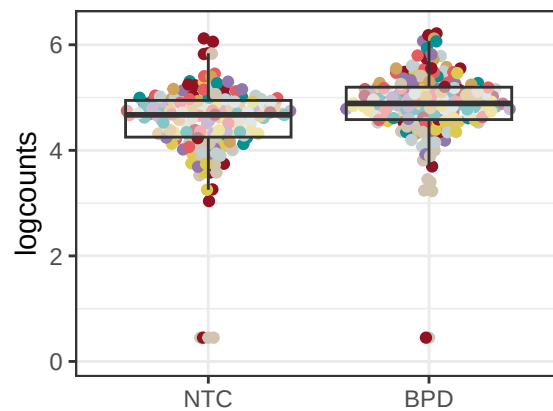
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}04$ logFC = 0.42



Seurat labels

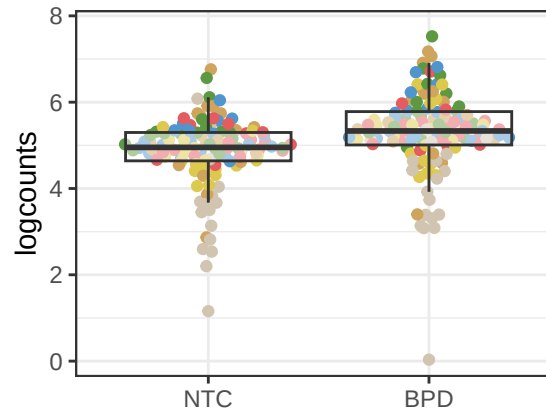
L-A sig: adj p < $3\text{e-}03$ logFC = 0.39



FOSL2

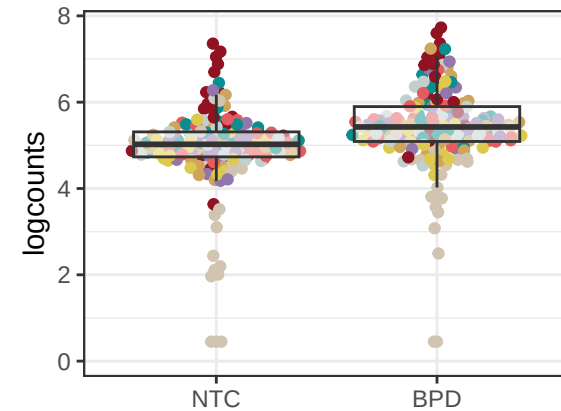
PRECAST (smoothed)

L-A sig: adj p < $5\text{e-}04$ logFC = 0.51



Seurat labels

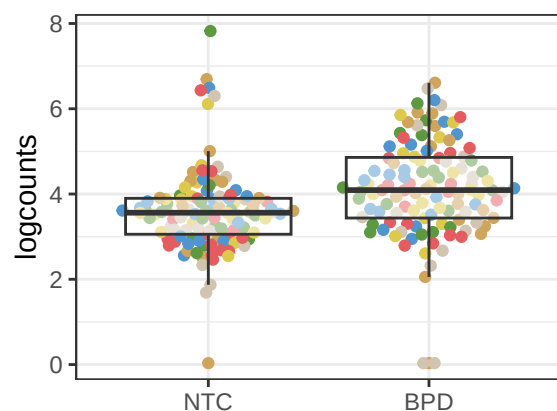
L-A sig: adj p < $2\text{e-}04$ logFC = 0.51



GADD45G

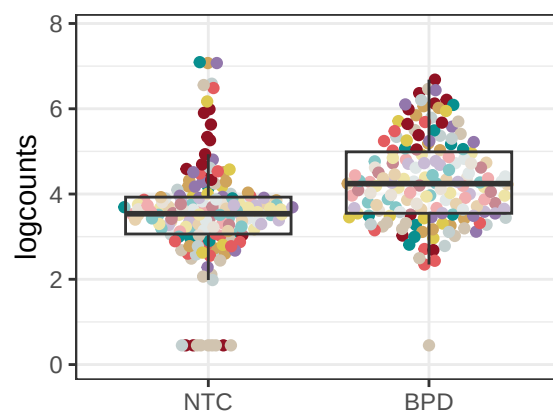
PRECAST (smoothed)

L-A sig: adj p < $1\text{e-}03$ logFC = 0.74



Seurat labels

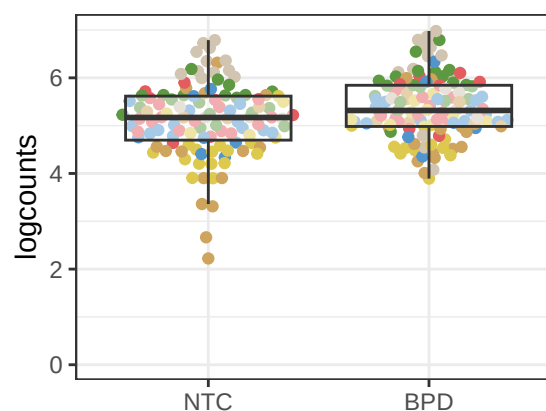
L-A sig: adj p < $3\text{e-}03$ logFC = 0.66



PSRC1

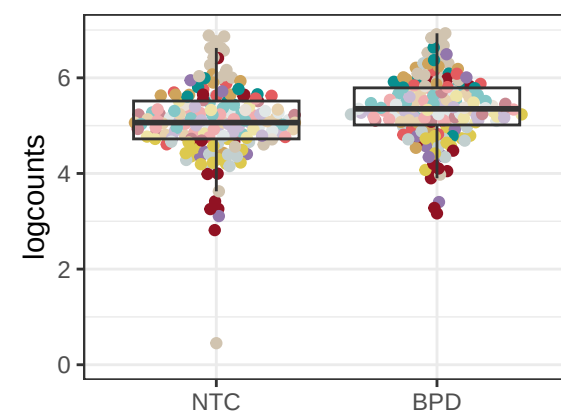
PRECAST (smoothed)

L-A sig: adj p < $1\text{e-}02$ logFC = 0.33



Seurat labels

L-A sig: adj p < $1\text{e-}02$ logFC = 0.33

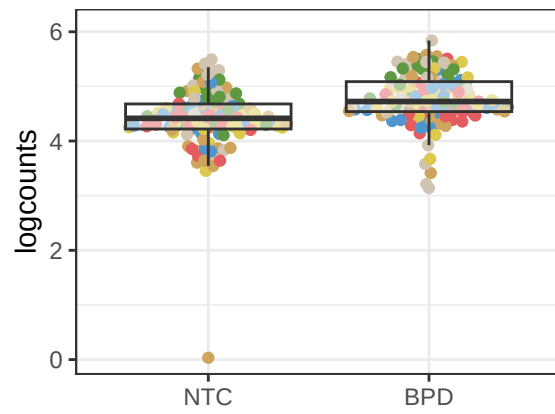


All genes with $\text{abs}(\log\text{FC}) > .5$

ELMSAN1

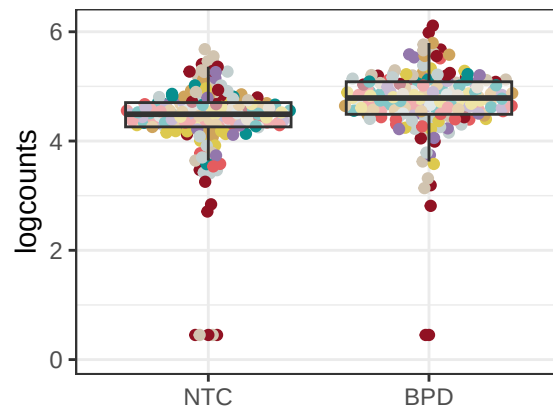
PRECAST (smoothed)

L-A sig: adj p < 2e-03 logFC = 0.35



Seurat labels

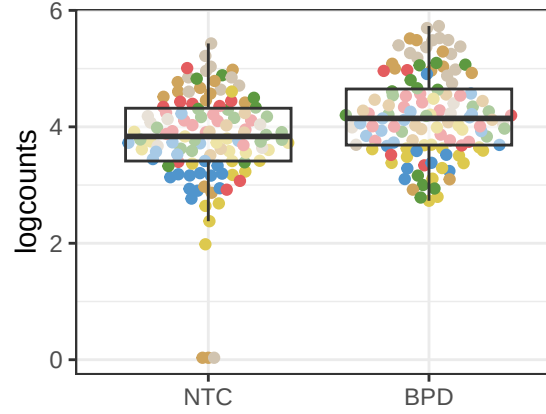
L-A sig: adj p < 3e-03 logFC = 0.37



BTG3

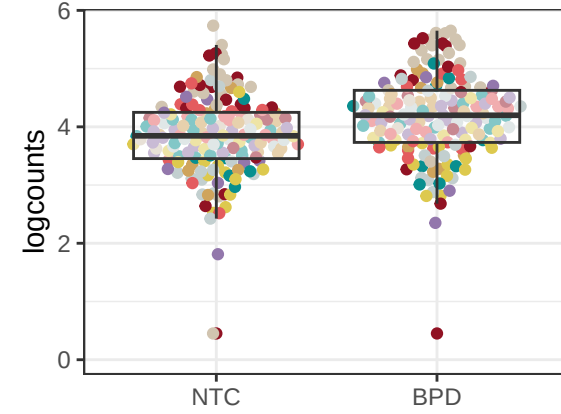
PRECAST (smoothed)

L-A sig: adj p < 3e-02 logFC = 0.37



Seurat labels

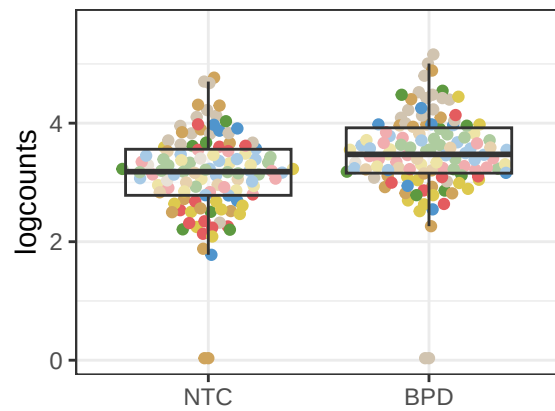
L-A sig: adj p < 3e-02 logFC = 0.38



EVA1C

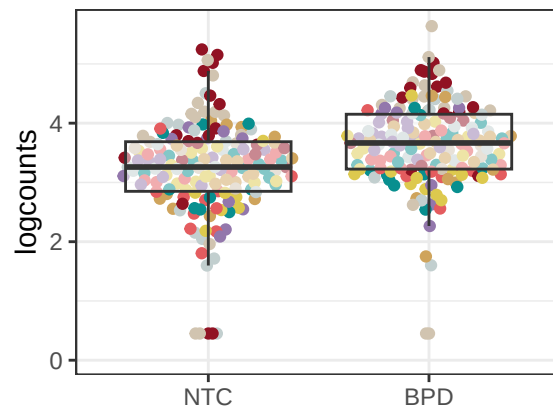
PRECAST (smoothed)

L-A sig: adj p < 8e-03 logFC = 0.44



Seurat labels

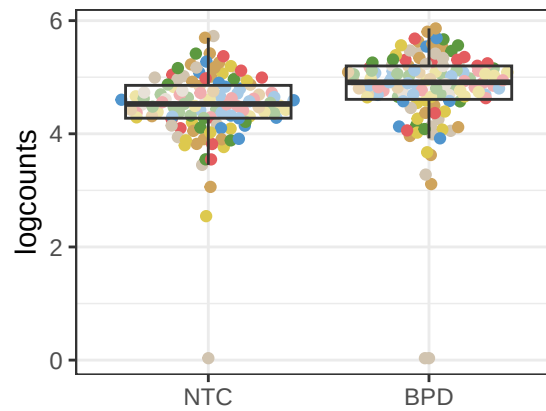
L-A sig: adj p < 4e-02 logFC = 0.4



SMAD1

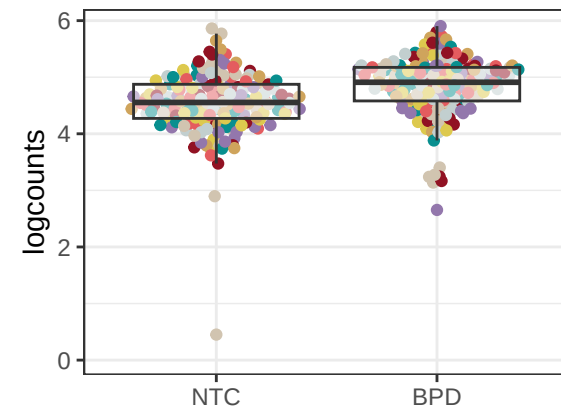
PRECAST (smoothed)

L-A sig: adj p < 7e-04 logFC = 0.36



Seurat labels

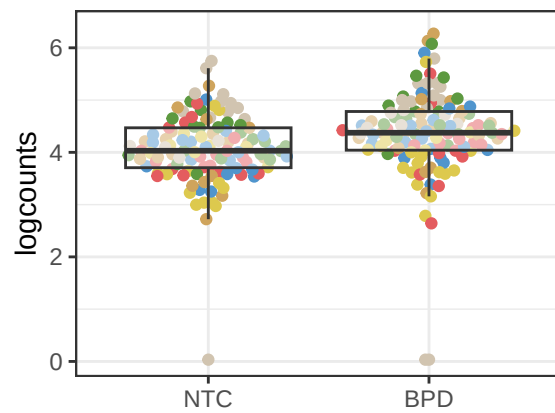
L-A sig: adj p < 7e-04 logFC = 0.37



ZNF395

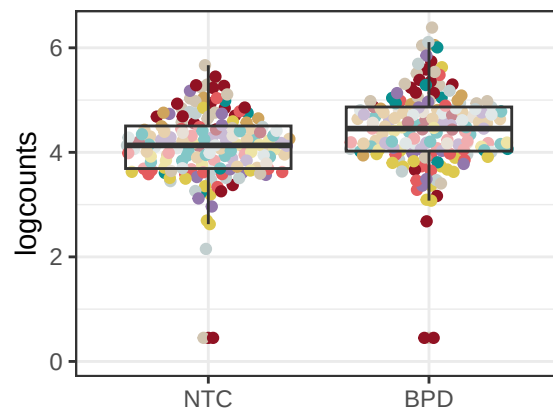
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = 0.35



Seurat labels

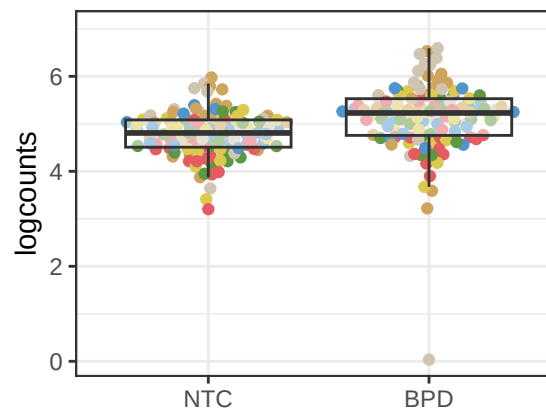
L-A sig: adj p < 4e-02 logFC = 0.36



HMGB2

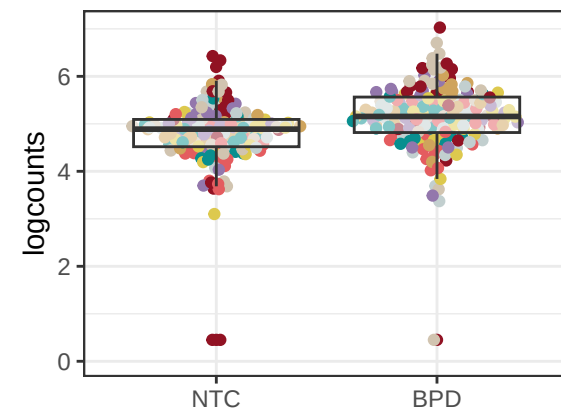
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = 0.3



Seurat labels

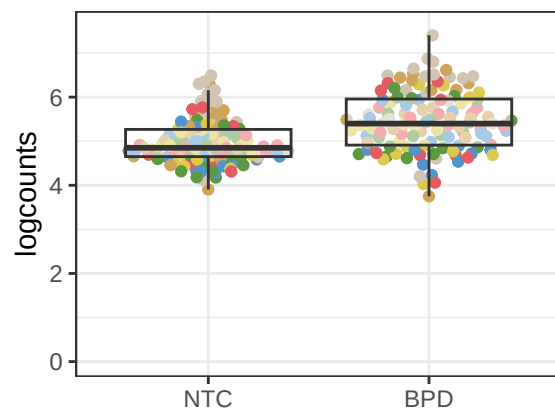
L-A sig: adj p < 2e-02 logFC = 0.31



ARRDC3

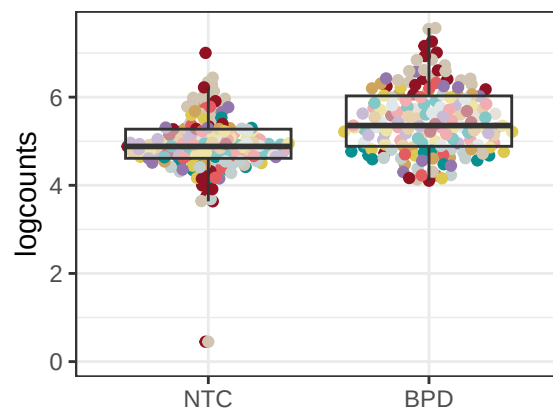
PRECAST (smoothed)

L-A sig: adj p < 3e-03 logFC = 0.42



Seurat labels

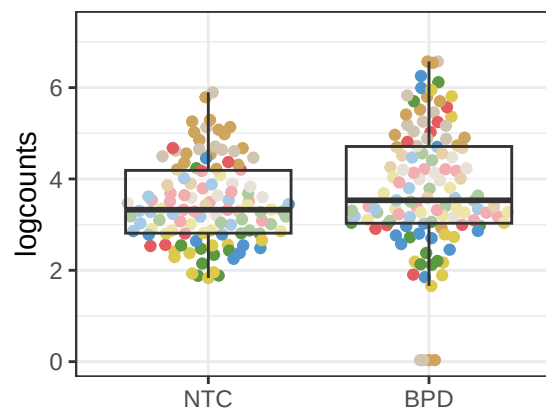
L-A sig: adj p < 1e-02 logFC = 0.39



ZIC1

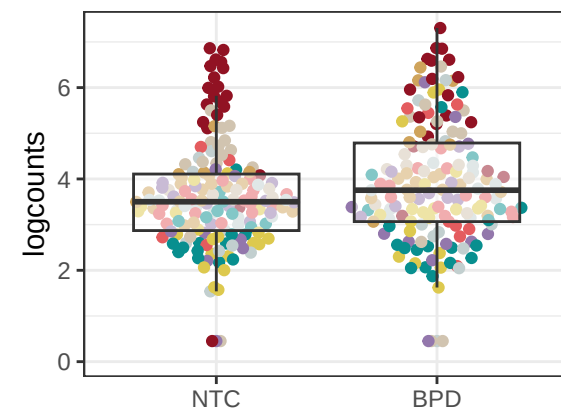
PRECAST (smoothed)

L-A sig: adj p < 4e-02 logFC = 0.51



Seurat labels

L-A sig: adj p < 3e-02 logFC = 0.53

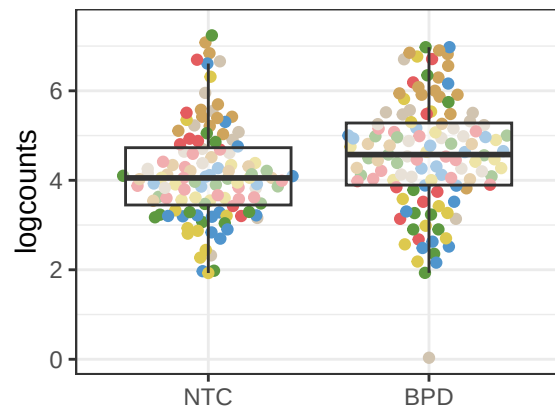


All genes with $\text{abs}(\log\text{FC}) > .5$

BAG3

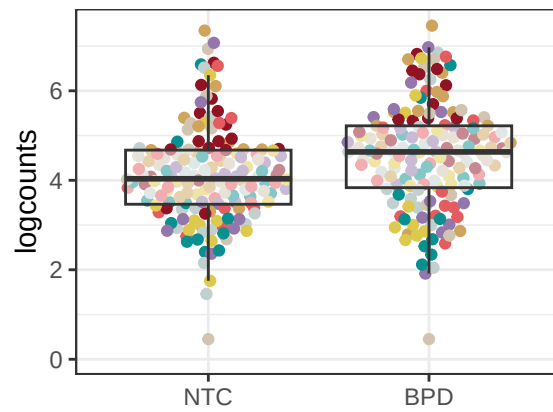
PRECAST (smoothed)

L-A sig: adj p < 4e-02 logFC = 0.6



Seurat labels

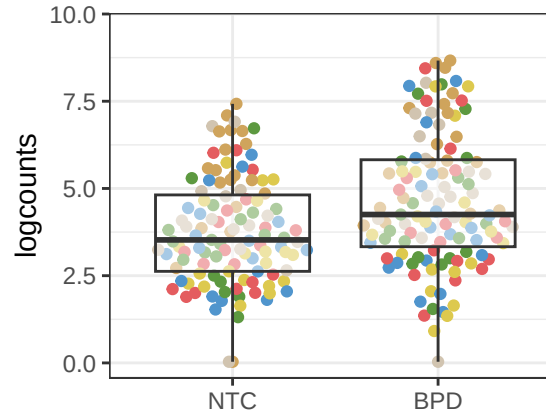
L-A sig: adj p < 1e-02 logFC = 0.65



ZFP36

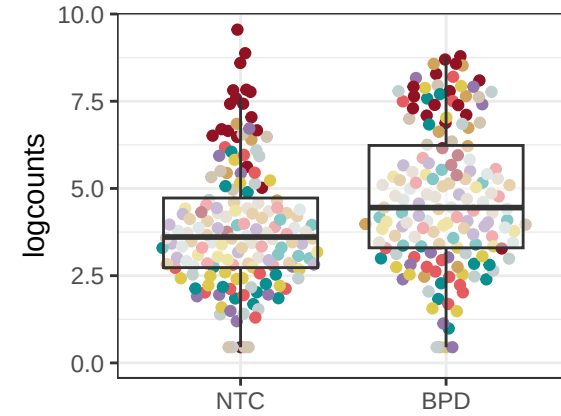
PRECAST (smoothed)

L-A sig: adj p < 1e-02 logFC = 1



Seurat labels

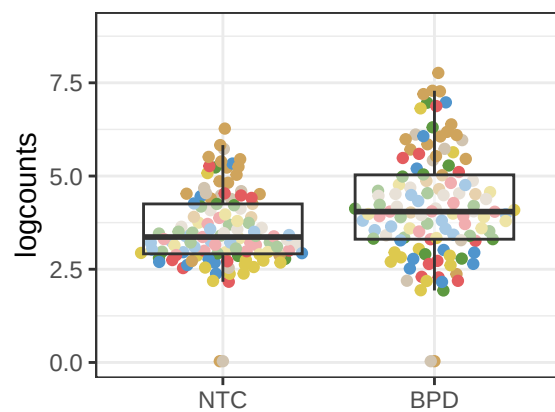
L-A sig: adj p < 9e-03 logFC = 0.91



YBX3

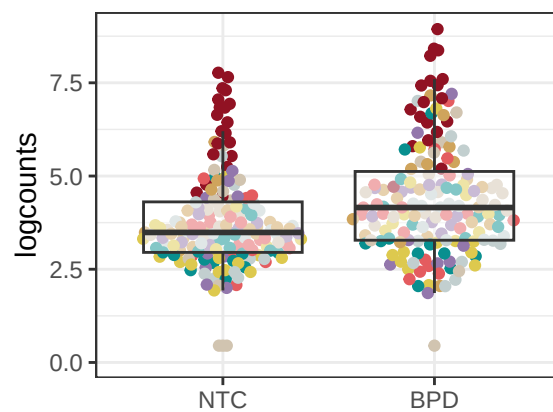
PRECAST (smoothed)

L-A sig: adj p < 1e-03 logFC = 0.81



Seurat labels

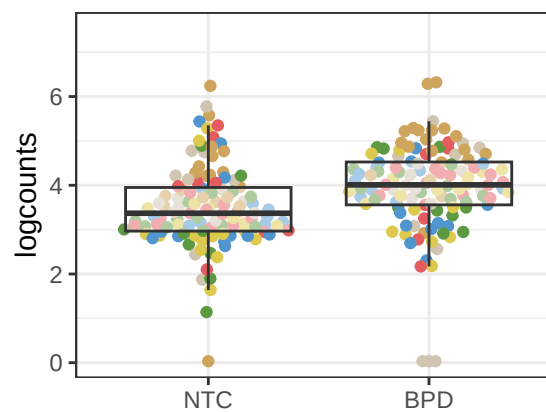
L-A sig: adj p < 4e-04 logFC = 0.79



CAVIN2

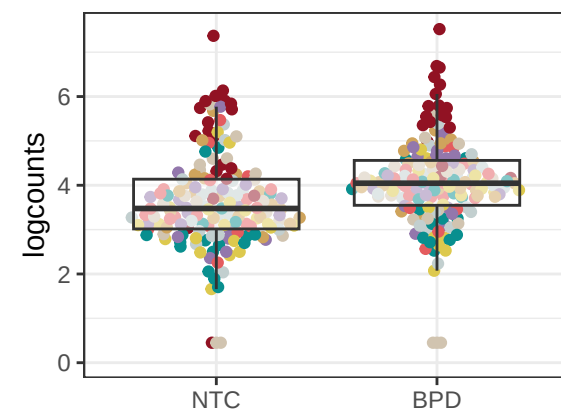
PRECAST (smoothed)

L-A sig: adj p < 4e-02 logFC = 0.43



Seurat labels

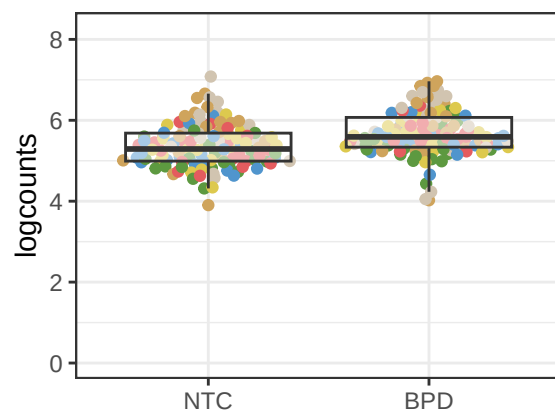
L-A sig: adj p < 1e-02 logFC = 0.46



NFKBIA

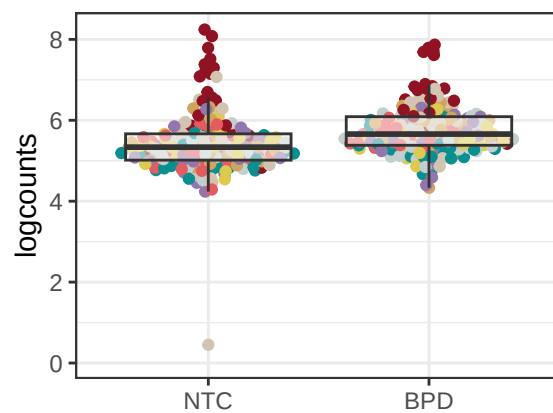
PRECAST (smoothed)

L-A sig: adj p < 1e-02 logFC = 0.33



Seurat labels

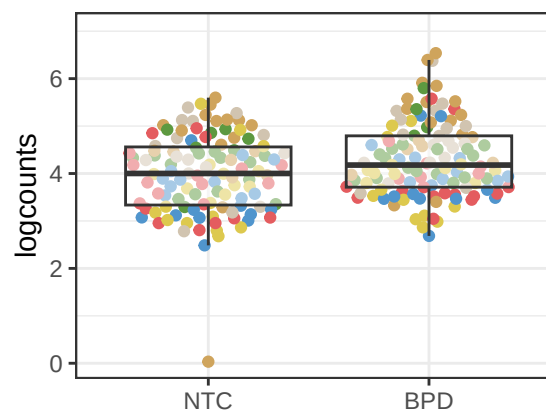
L-A sig: adj p < 1e-02 logFC = 0.32



TNFRSF1A

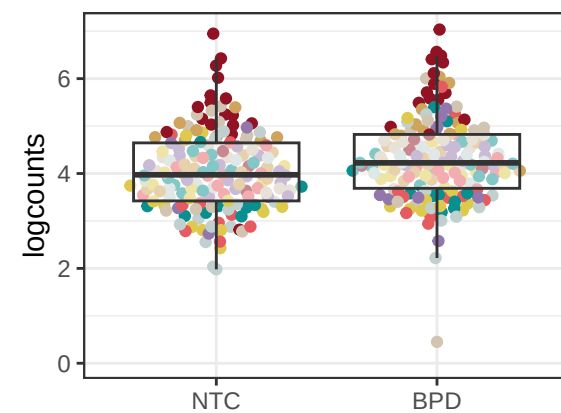
PRECAST (smoothed)

L-A sig: adj p < 4e-02 logFC = 0.4



Seurat labels

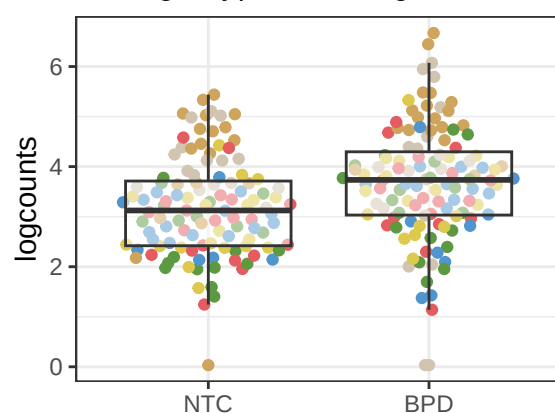
L-A sig: adj p < 2e-02 logFC = 0.42



PDLIM4

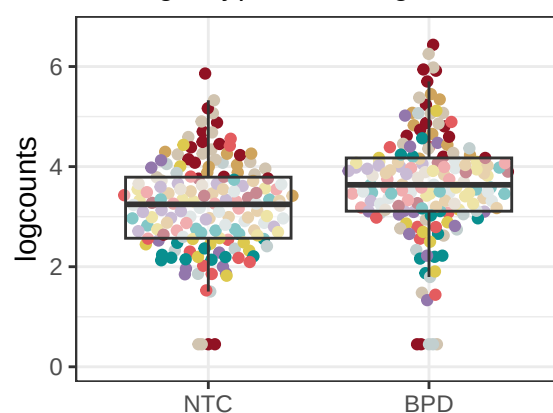
PRECAST (smoothed)

L-A sig: adj p < 1e-04 logFC = 0.84



Seurat labels

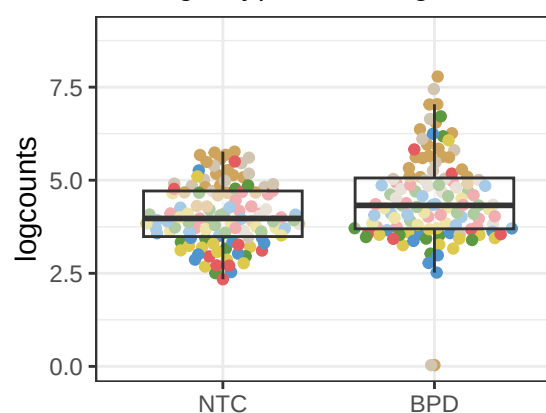
L-A sig: adj p < 4e-05 logFC = 0.85



TIMP1

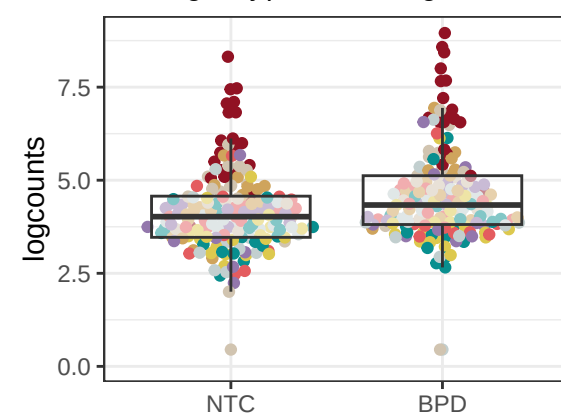
PRECAST (smoothed)

L-A sig: adj p < 4e-03 logFC = 0.58



Seurat labels

L-A sig: adj p < 5e-03 logFC = 0.55

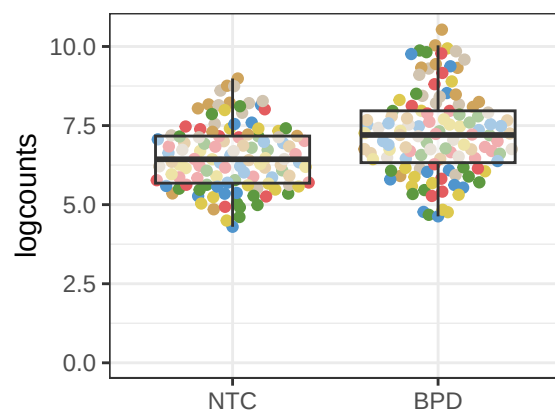


All genes with $\text{abs}(\log\text{FC}) > .5$

MT1X

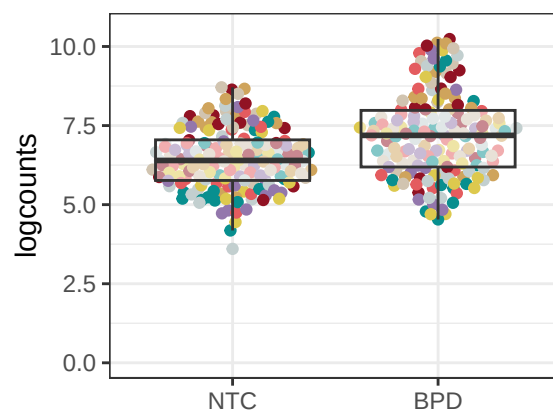
PRECAST (smoothed)

L-A sig: adj p < $3\text{e-}04$ logFC = 0.82



Seurat labels

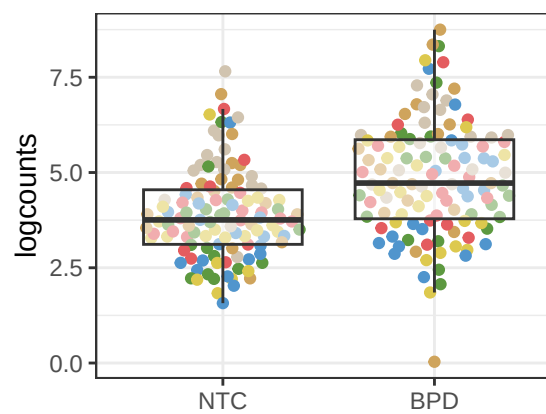
L-A sig: adj p < $2\text{e-}05$ logFC = 0.83



ANGPTL4

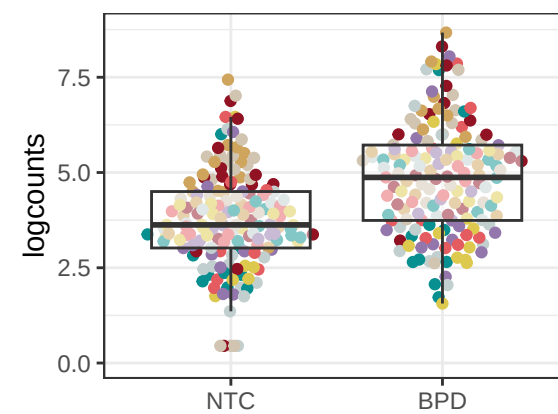
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}04$ logFC = 1.14



Seurat labels

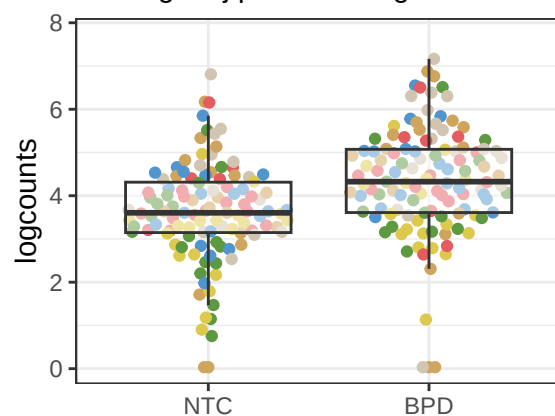
L-A sig: adj p < $3\text{e-}05$ logFC = 1.15



FKBP5

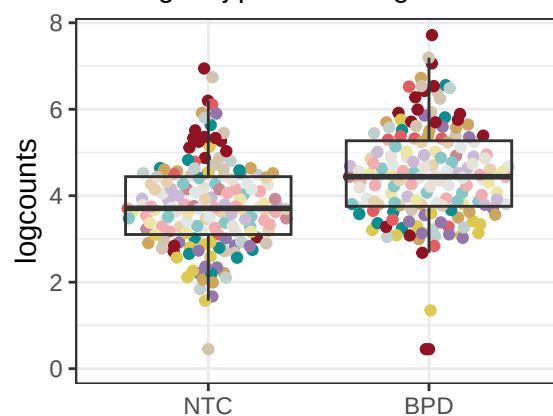
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}02$ logFC = 0.64



Seurat labels

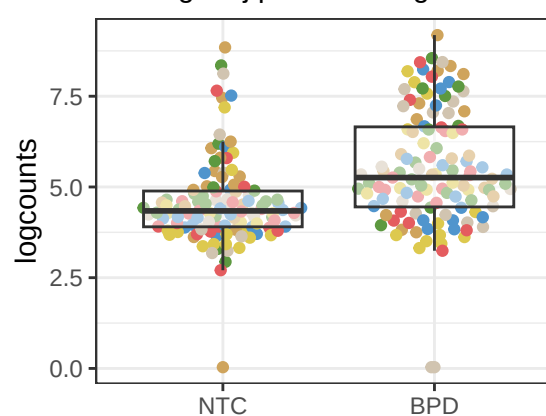
L-A sig: adj p < $8\text{e-}03$ logFC = 0.62



GADD45B

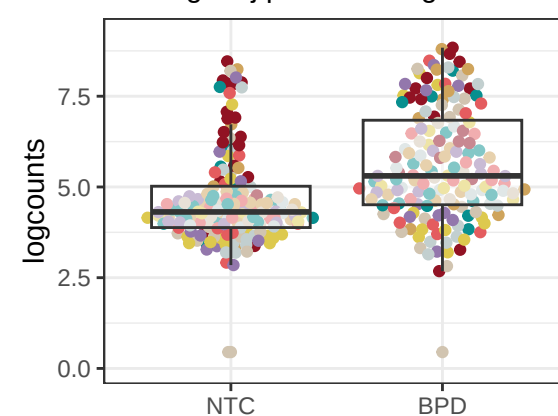
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}05$ logFC = 1.16



Seurat labels

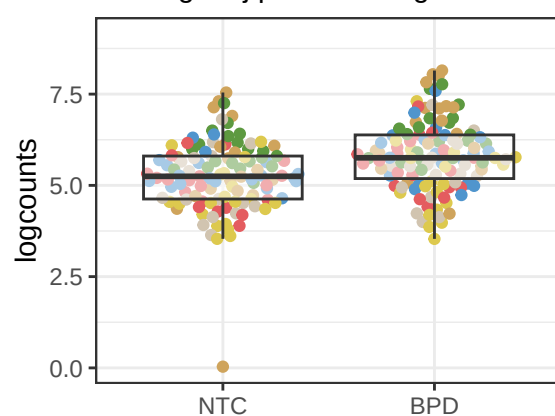
L-A sig: adj p < $1\text{e-}07$ logFC = 1.21



CEBPD

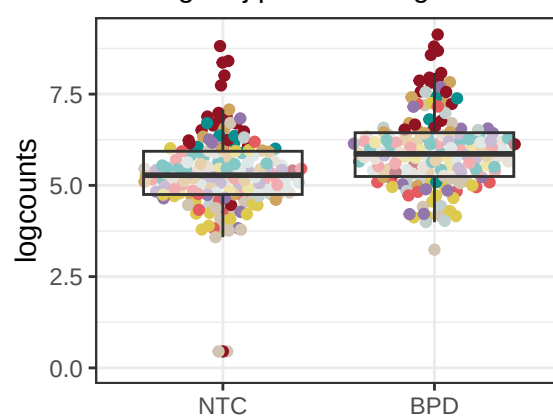
PRECAST (smoothed)

L-A sig: adj p < $1\text{e-}03$ logFC = 0.58



Seurat labels

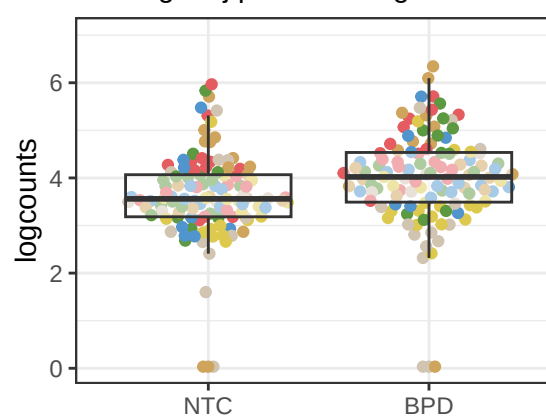
L-A sig: adj p < $1\text{e-}04$ logFC = 0.59



ARID5A

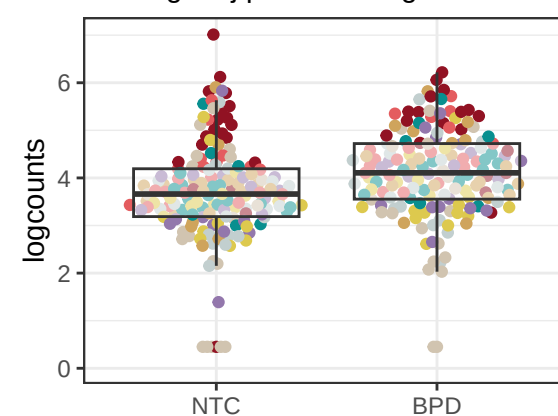
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}03$ logFC = 0.55



Seurat labels

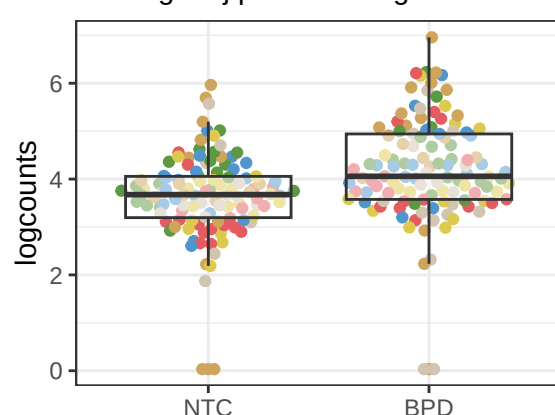
L-A sig: adj p < $8\text{e-}04$ logFC = 0.57



VASN

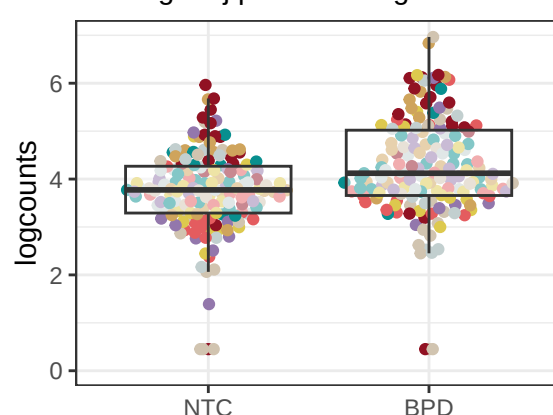
PRECAST (smoothed)

L-A sig: adj p < $5\text{e-}04$ logFC = 0.59



Seurat labels

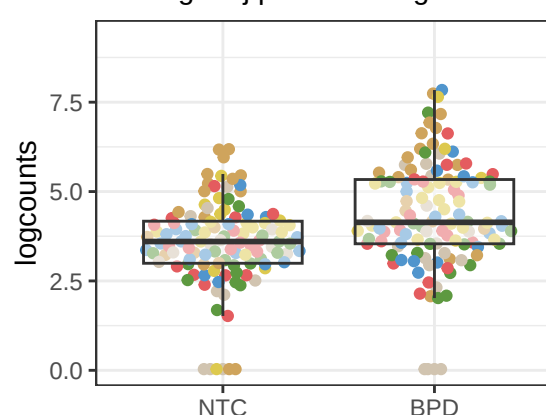
L-A sig: adj p < $1\text{e-}04$ logFC = 0.63



DEPP1

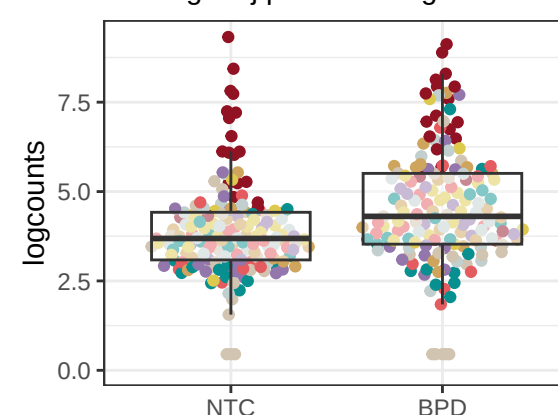
PRECAST (smoothed)

L-A sig: adj p < $1\text{e-}02$ logFC = 0.77



Seurat labels

L-A sig: adj p < $2\text{e-}03$ logFC = 0.82

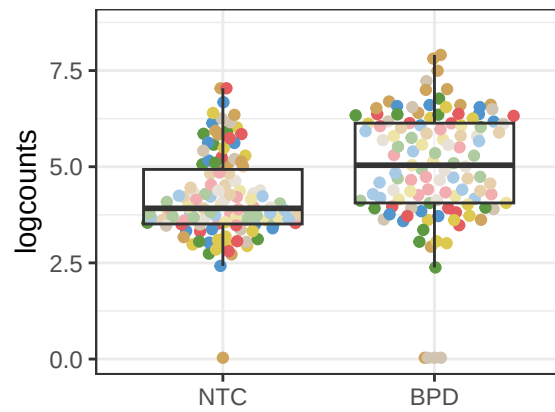


All genes with $\text{abs}(\log\text{FC}) > .5$

APOLD1

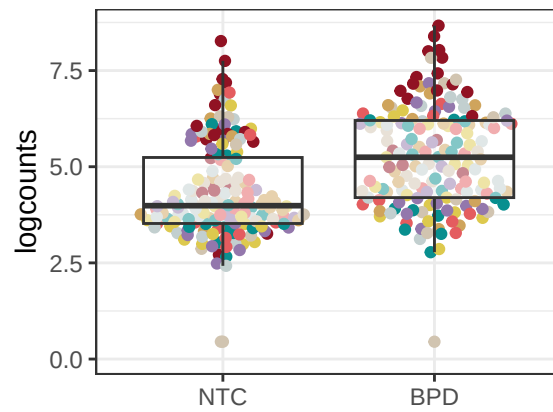
PRECAST (smoothed)

L-A sig: adj $p < 2\text{e-}05$ logFC = 0.95



Seurat labels

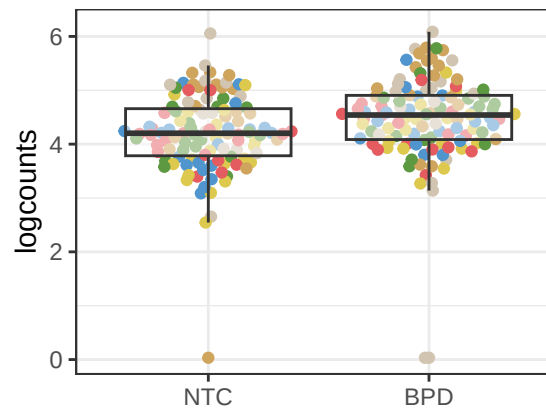
L-A sig: adj $p < 2\text{e-}06$ logFC = 0.9



SEMA4B

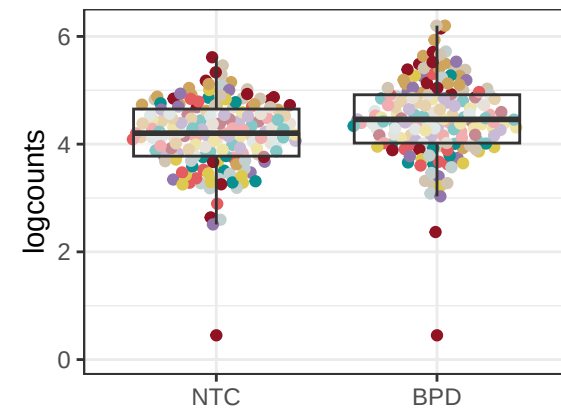
PRECAST (smoothed)

L-A sig: adj $p < 2\text{e-}02$ logFC = 0.38



Seurat labels

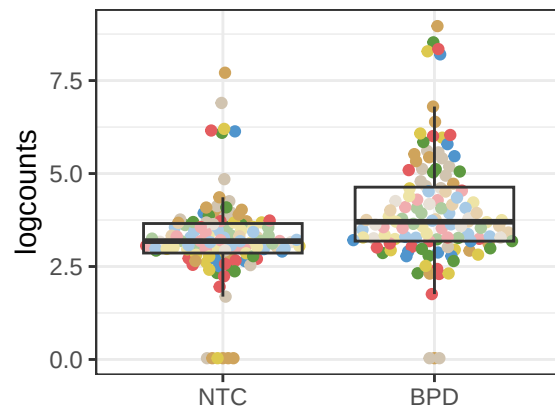
L-A sig: adj $p < 3\text{e-}02$ logFC = 0.35



HILPDA

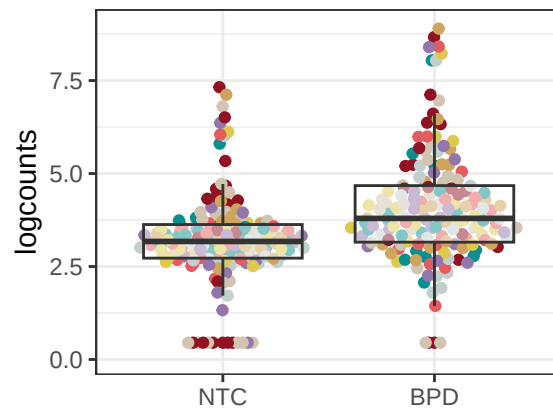
PRECAST (smoothed)

L-A sig: adj $p < 2\text{e-}02$ logFC = 0.78



Seurat labels

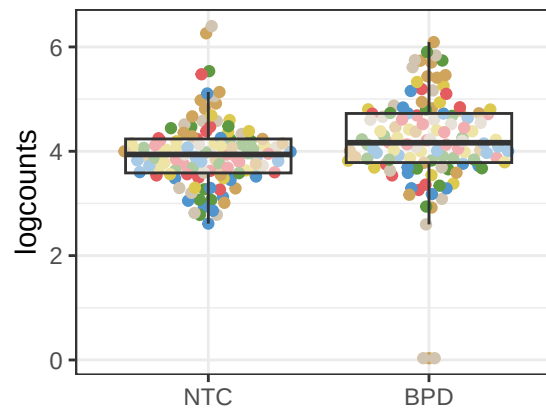
L-A sig: adj $p < 6\text{e-}03$ logFC = 0.79



CSRNP1

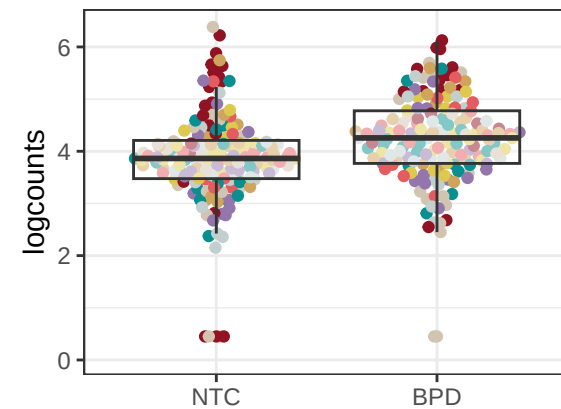
PRECAST (smoothed)

L-A sig: adj $p < 3\text{e-}02$ logFC = 0.47



Seurat labels

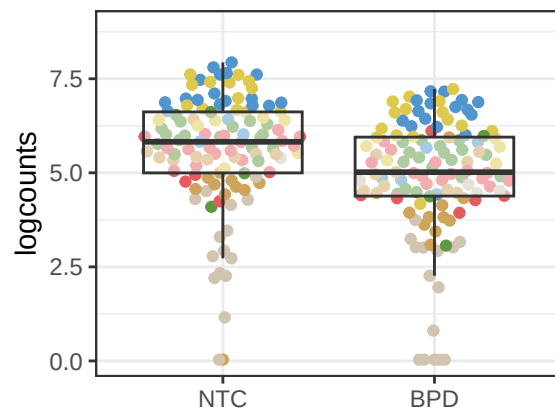
L-A sig: adj $p < 3\text{e-}02$ logFC = 0.46



PVALB

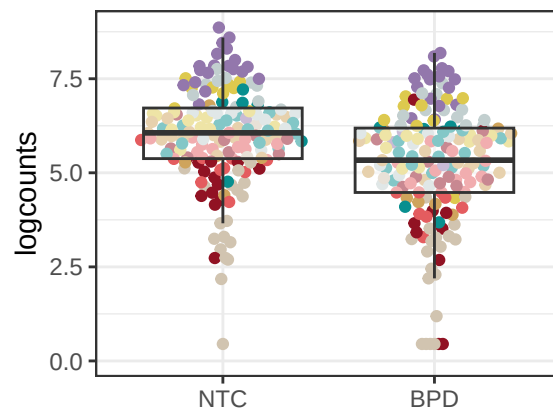
PRECAST (smoothed)

L-A sig: adj $p < 4\text{e-}04$ logFC = -0.65



Seurat labels

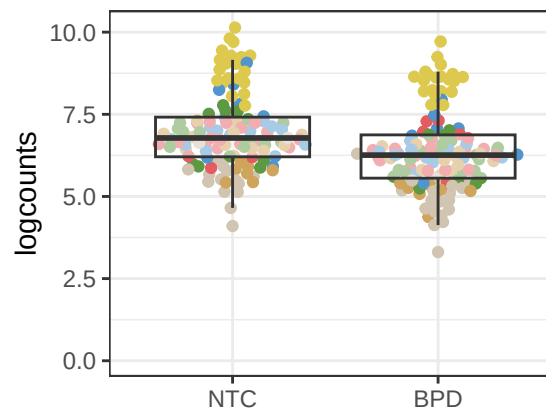
L-A sig: adj $p < 6\text{e-}04$ logFC = -0.63



PCP4

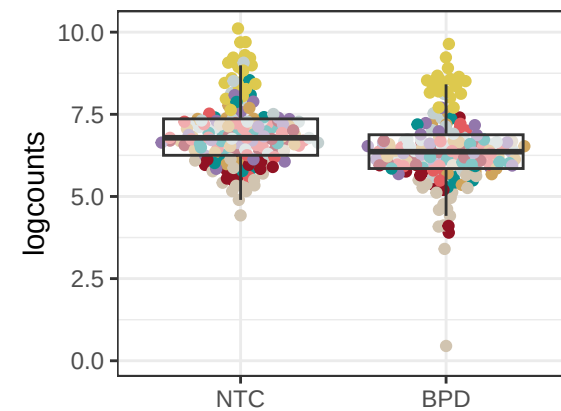
PRECAST (smoothed)

L-A sig: adj $p < 5\text{e-}03$ logFC = -0.4



Seurat labels

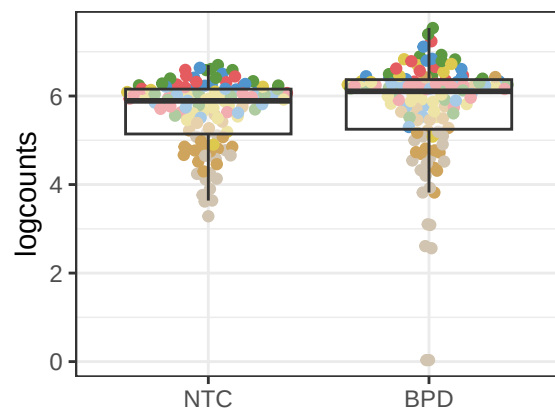
L-A sig: adj $p < 7\text{e-}04$ logFC = -0.4



RTN4RL2

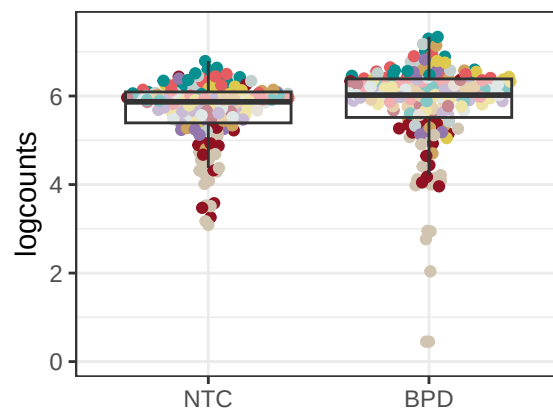
PRECAST (smoothed)

L-A sig: adj $p < 2\text{e-}02$ logFC = 0.31



Seurat labels

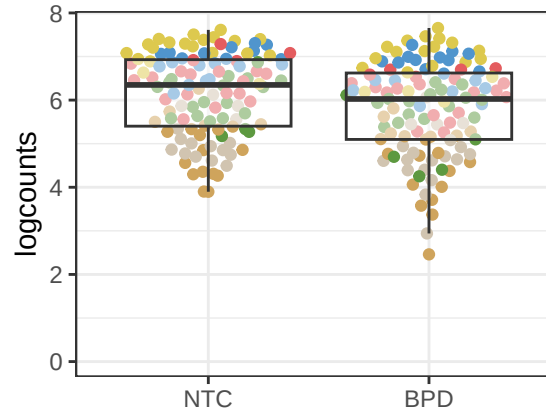
L-A sig: adj $p < 2\text{e-}02$ logFC = 0.31



NEUROD6

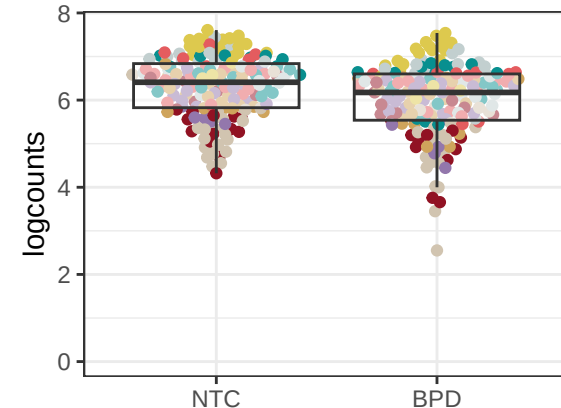
PRECAST (smoothed)

L-A sig: adj $p < 3\text{e-}02$ logFC = -0.34



Seurat labels

L-A sig: adj $p < 9\text{e-}03$ logFC = -0.34

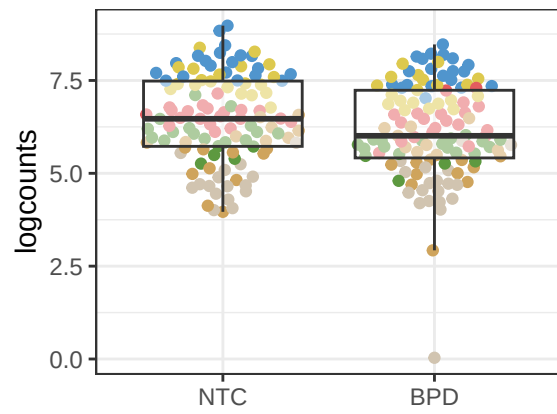


All genes with $\text{abs}(\log\text{FC}) > .5$

NEFH

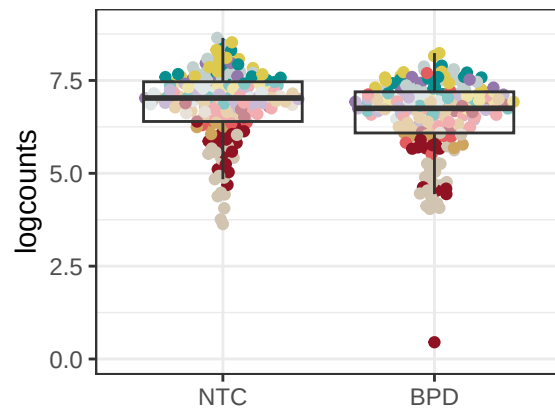
PRECAST (smoothed)

L-A sig: adj p < 5e-02 logFC = -0.31



Seurat labels

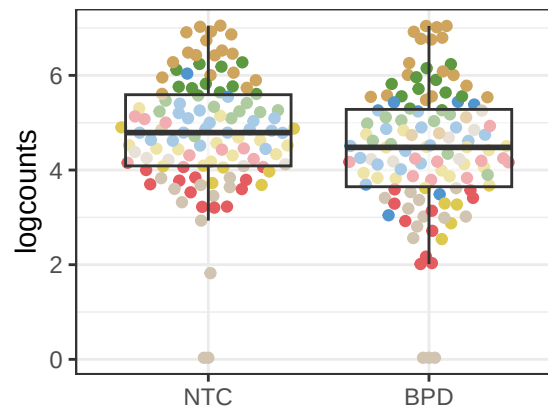
L-A sig: adj p < 2e-02 logFC = -0.33



DLX6-AS1

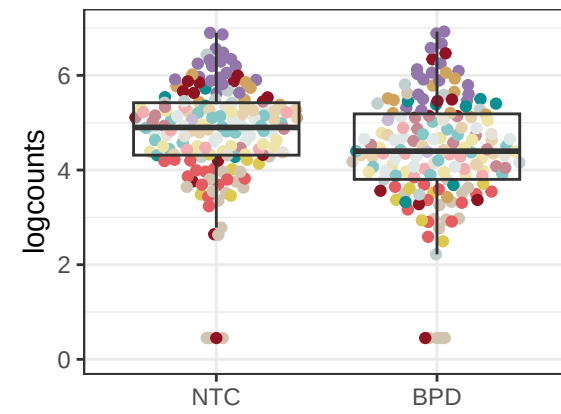
PRECAST (smoothed)

L-A sig: adj p < 2e-04 logFC = -0.44



Seurat labels

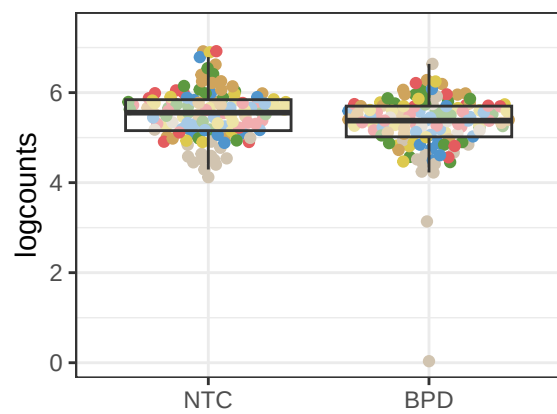
L-A sig: adj p < 7e-03 logFC = -0.39



ZNF385D

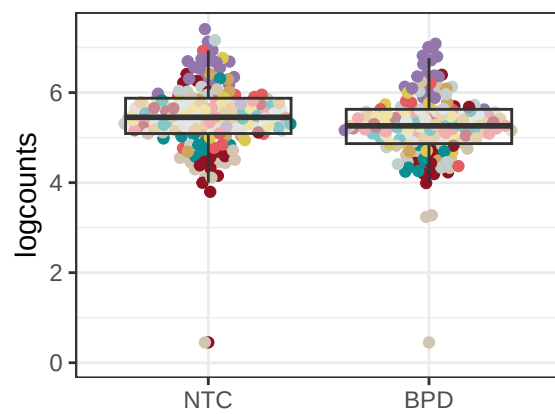
PRECAST (smoothed)

L-A sig: adj p < 2e-03 logFC = -0.32



Seurat labels

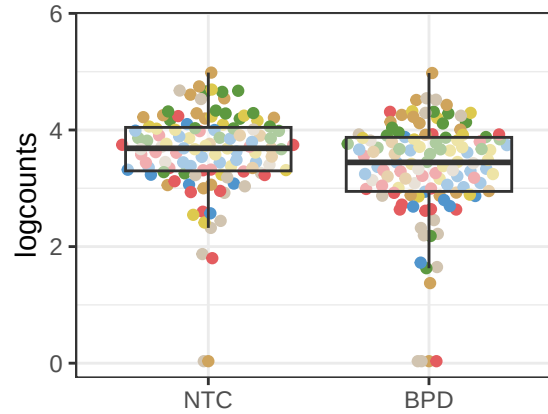
L-A sig: adj p < 4e-03 logFC = -0.33



AC067773.1

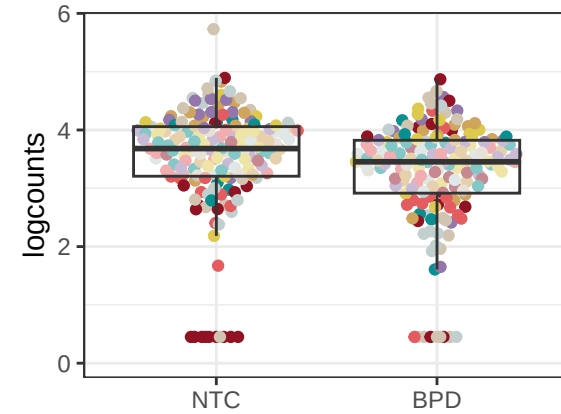
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = -0.41



Seurat labels

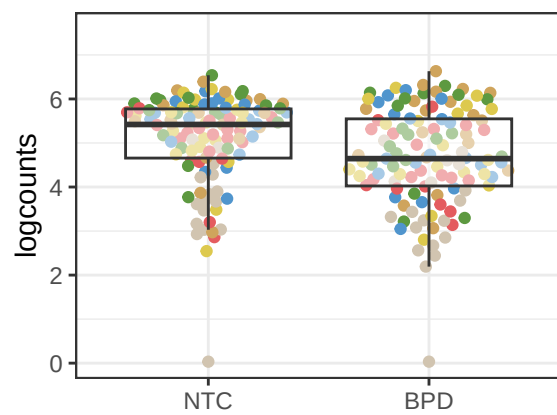
L-A sig: adj p < 3e-02 logFC = -0.41



SLC32A1

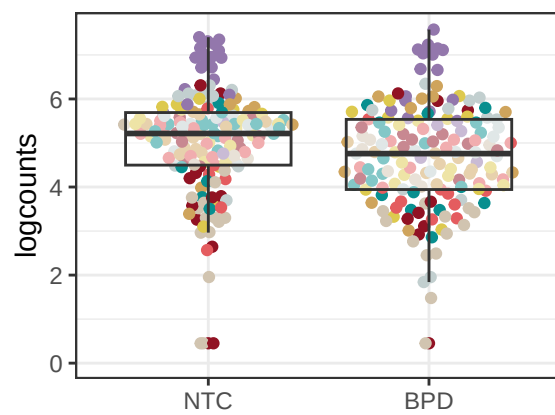
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = -0.49



Seurat labels

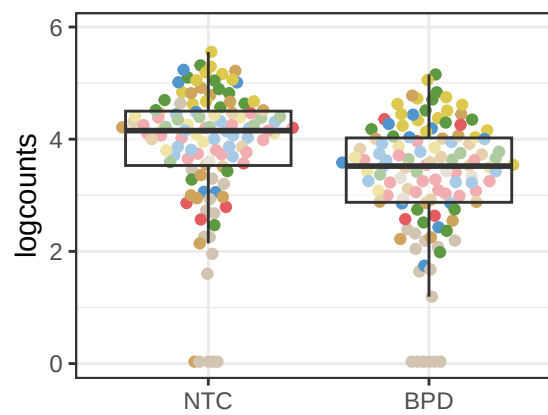
L-A sig: adj p < 3e-02 logFC = -0.49



TRBC2

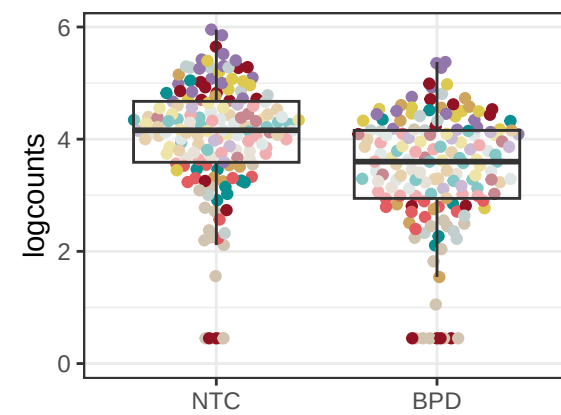
PRECAST (smoothed)

L-A sig: adj p < 4e-03 logFC = -0.56



Seurat labels

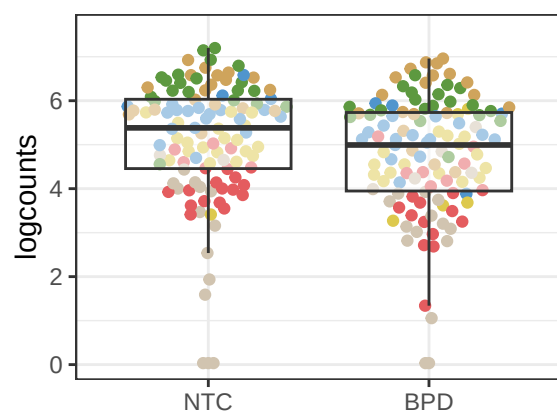
L-A sig: adj p < 2e-02 logFC = -0.48



VIP

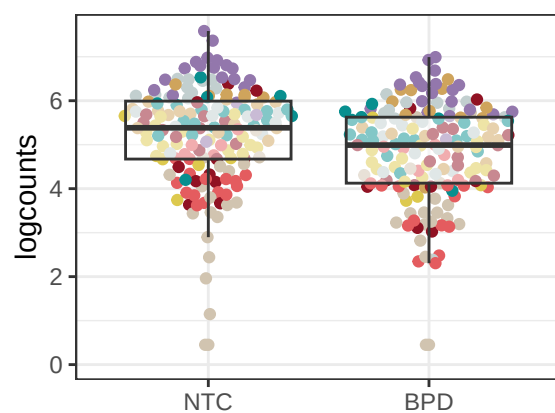
PRECAST (smoothed)

L-A sig: adj p < 2e-03 logFC = -0.45



Seurat labels

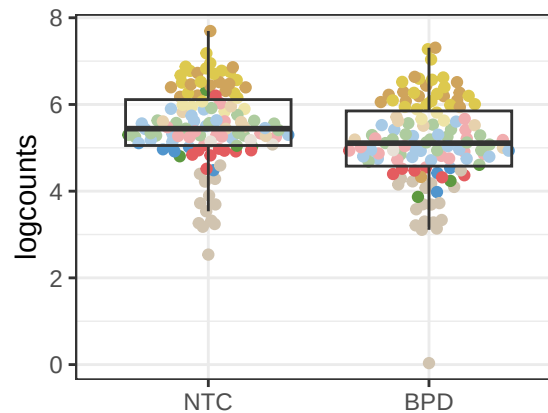
L-A sig: adj p < 2e-02 logFC = -0.42



PCP4L1

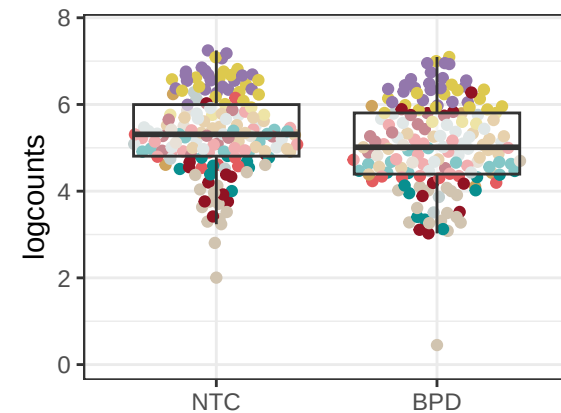
PRECAST (smoothed)

L-A sig: adj p < 2e-02 logFC = -0.3



Seurat labels

L-A sig: adj p < 2e-02 logFC = -0.32

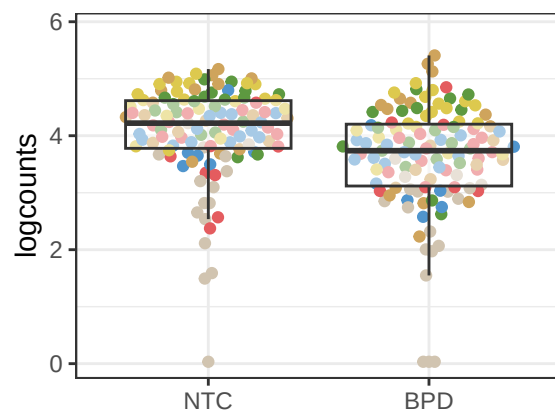


All genes with $\text{abs}(\log\text{FC}) > .5$

NXPH2

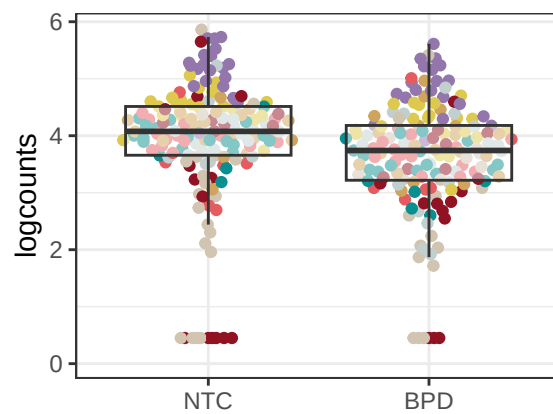
PRECAST (smoothed)

L-A sig: adj p < $3\text{e-}02$ logFC = -0.38



Seurat labels

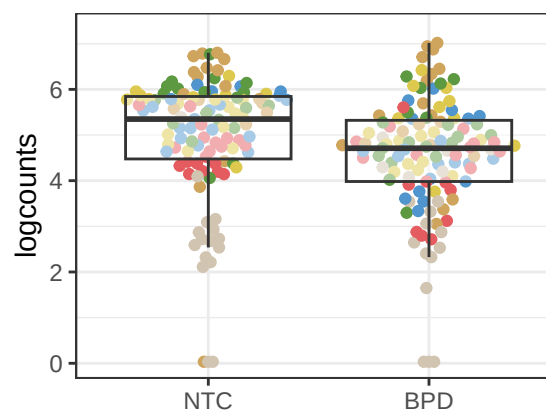
L-A sig: adj p < $3\text{e-}02$ logFC = -0.4



CRH

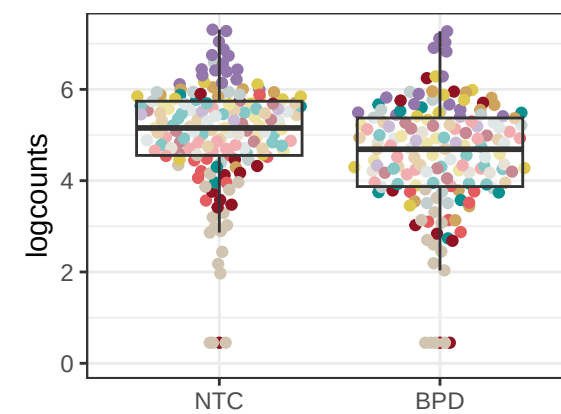
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}04$ logFC = -0.61



Seurat labels

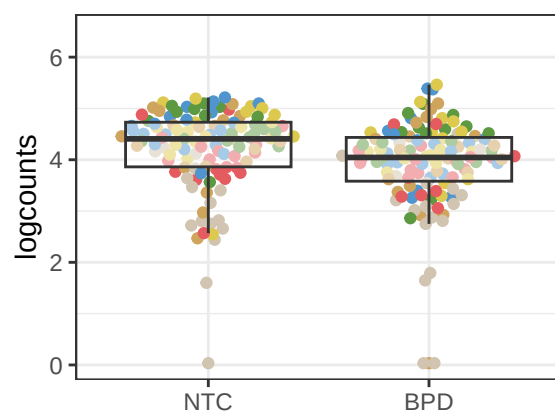
L-A sig: adj p < $1\text{e-}03$ logFC = -0.56



LGI2

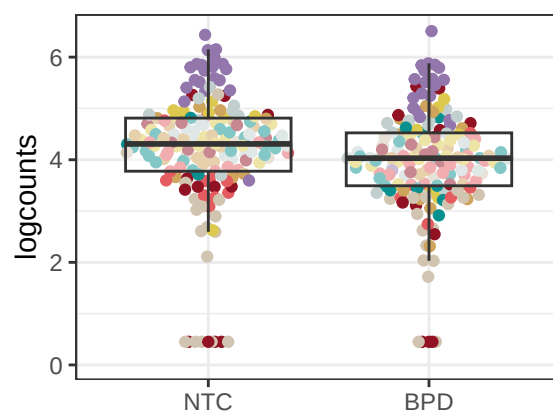
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}02$ logFC = -0.38



Seurat labels

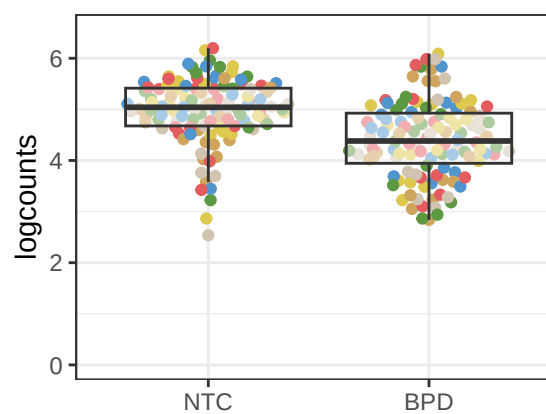
L-A sig: adj p < $4\text{e-}02$ logFC = -0.4



SLC35E2B

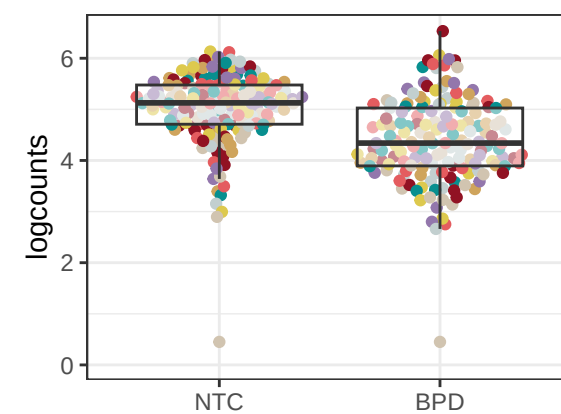
PRECAST (smoothed)

L-A sig: adj p < $5\text{e-}05$ logFC = -0.67



Seurat labels

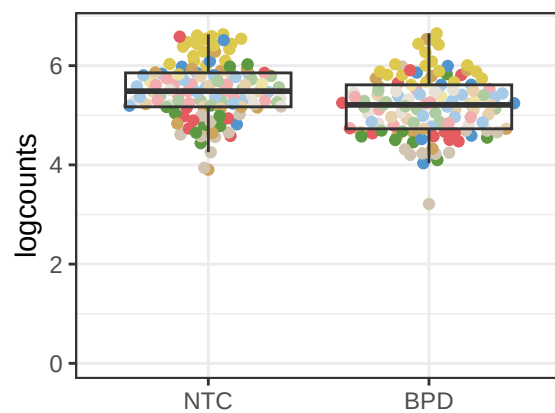
L-A sig: adj p < $3\text{e-}05$ logFC = -0.67



TRIB2

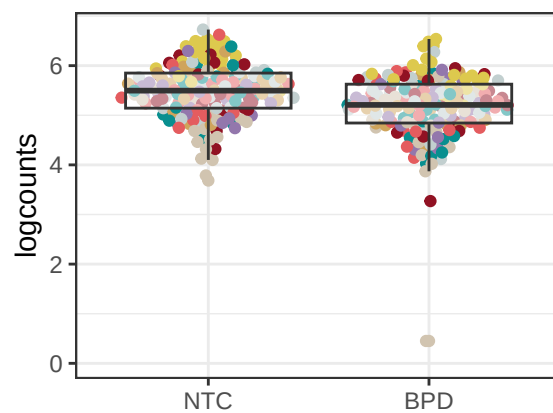
PRECAST (smoothed)

L-A sig: adj p < $4\text{e-}04$ logFC = -0.41



Seurat labels

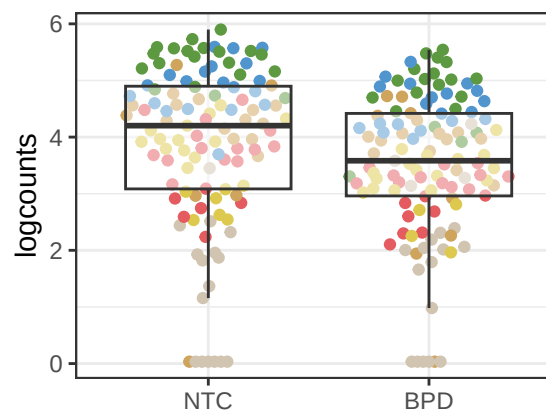
L-A sig: adj p < $6\text{e-}04$ logFC = -0.41



FBLN7

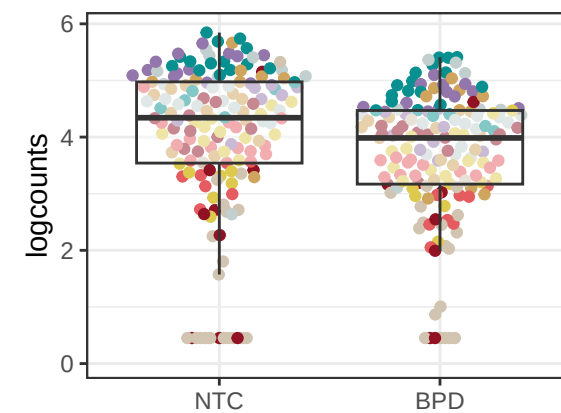
PRECAST (smoothed)

L-A sig: adj p < $1\text{e-}02$ logFC = -0.45



Seurat labels

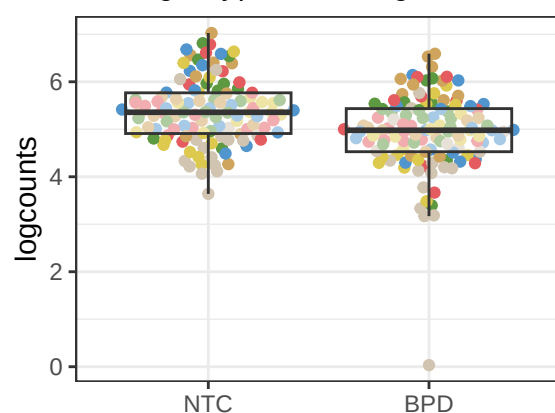
L-A sig: adj p < $1\text{e-}02$ logFC = -0.45



COL5A3

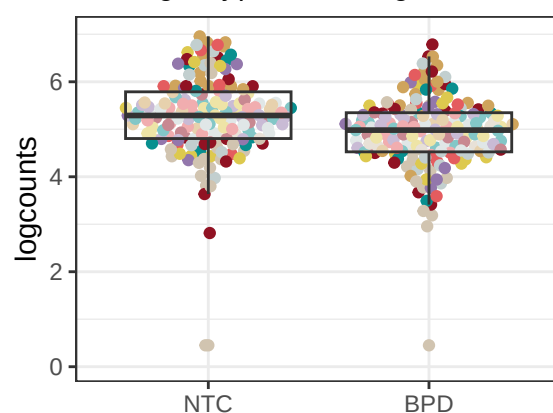
PRECAST (smoothed)

L-A sig: adj p < $8\text{e-}03$ logFC = -0.42



Seurat labels

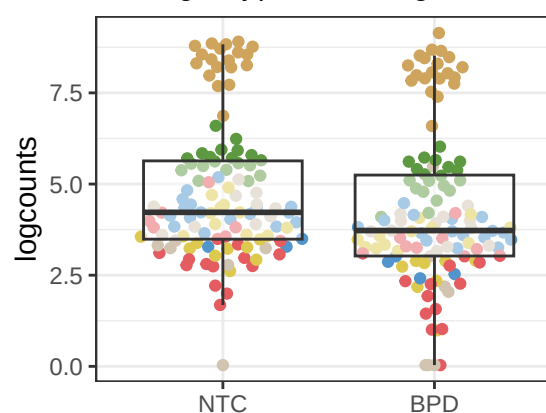
L-A sig: adj p < $1\text{e-}03$ logFC = -0.47



RELN

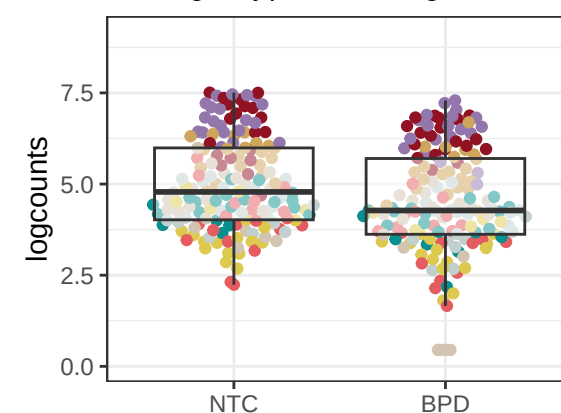
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}02$ logFC = -0.43



Seurat labels

L-A sig: adj p < $3\text{e-}02$ logFC = -0.4

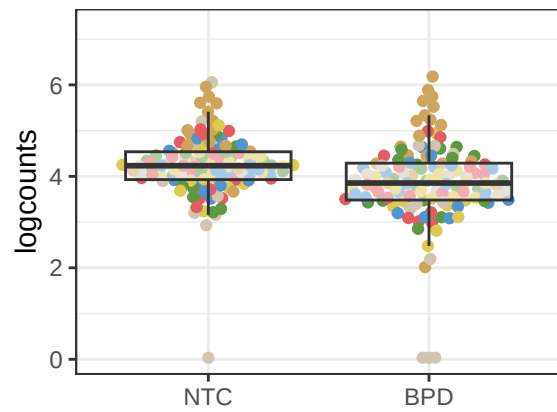


All genes with $\text{abs}(\log\text{FC}) > .5$

ITIH5

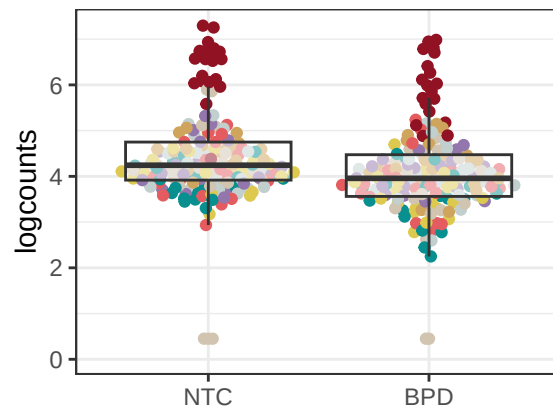
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}03$ logFC = -0.45



Seurat labels

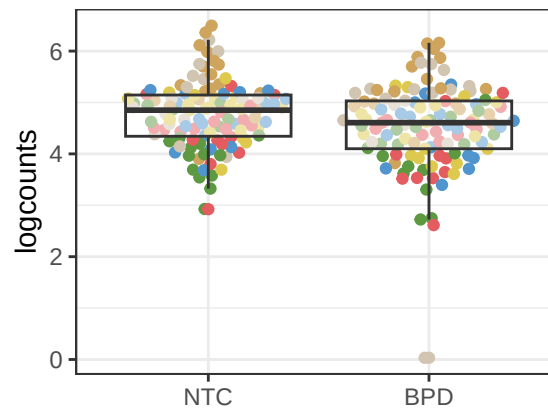
L-A sig: adj p < $2\text{e-}03$ logFC = -0.46



ASCL1

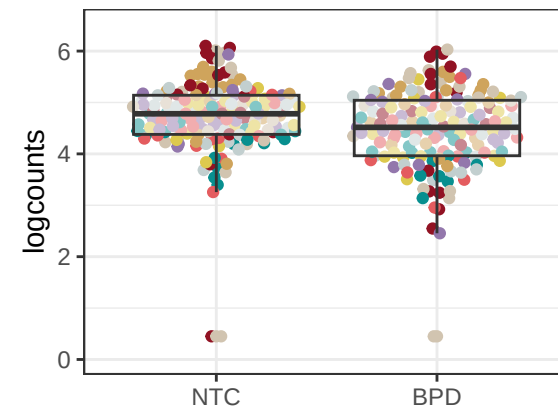
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}02$ logFC = -0.35



Seurat labels

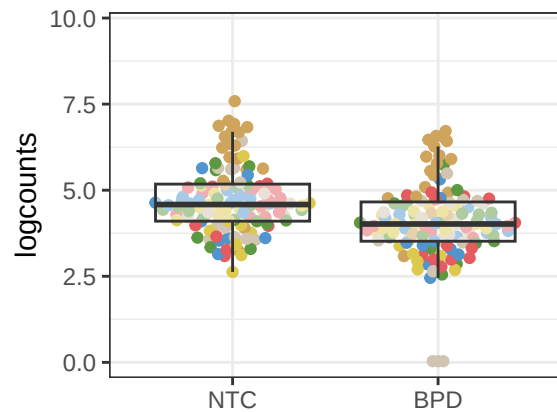
L-A sig: adj p < $1\text{e-}02$ logFC = -0.38



MUSTN1

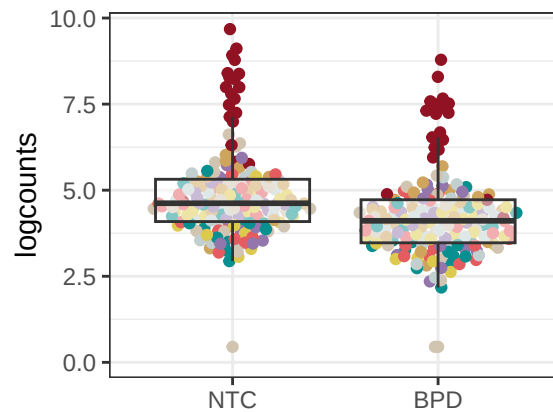
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}04$ logFC = -0.61



Seurat labels

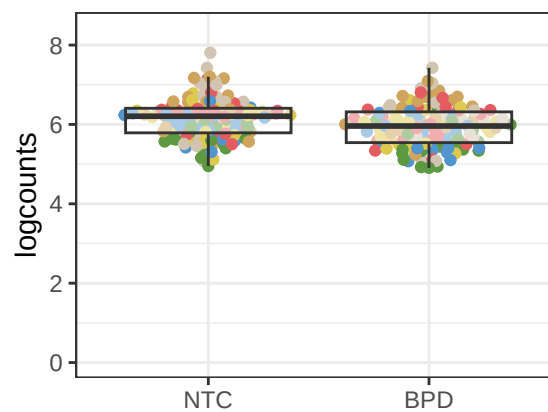
L-A sig: adj p < $2\text{e-}05$ logFC = -0.7



ITM2A

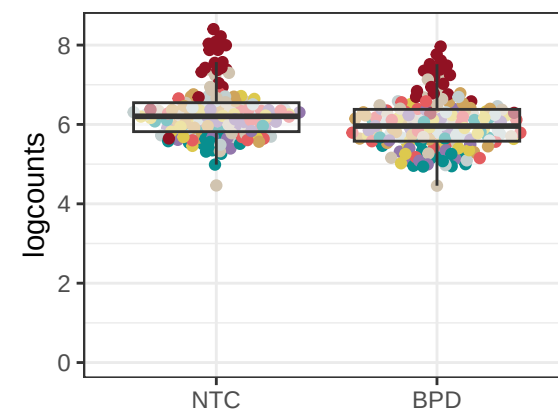
PRECAST (smoothed)

L-A sig: adj p < $4\text{e-}03$ logFC = -0.35



Seurat labels

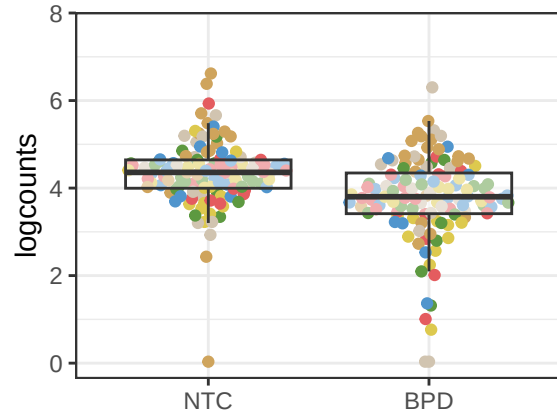
L-A sig: adj p < $4\text{e-}04$ logFC = -0.36



ABCG2

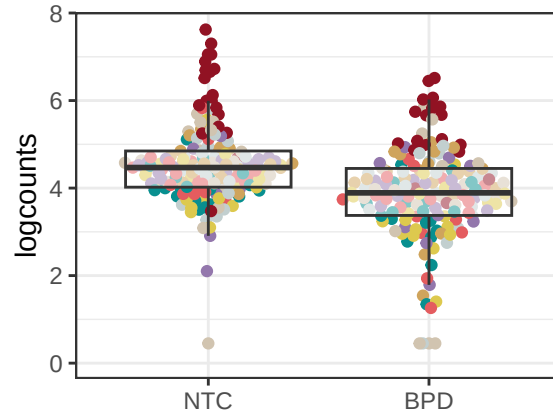
PRECAST (smoothed)

L-A sig: adj p < $4\text{e-}06$ logFC = -0.79



Seurat labels

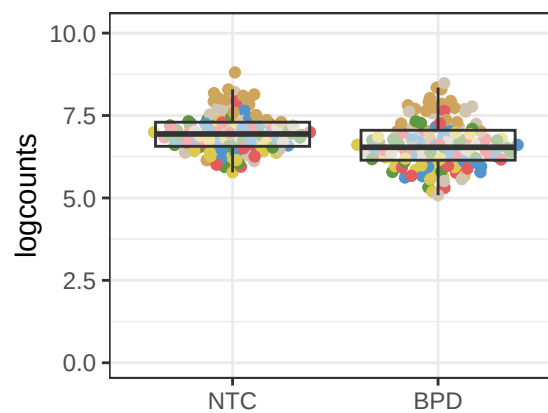
L-A sig: adj p < $7\text{e-}07$ logFC = -0.77



A2M

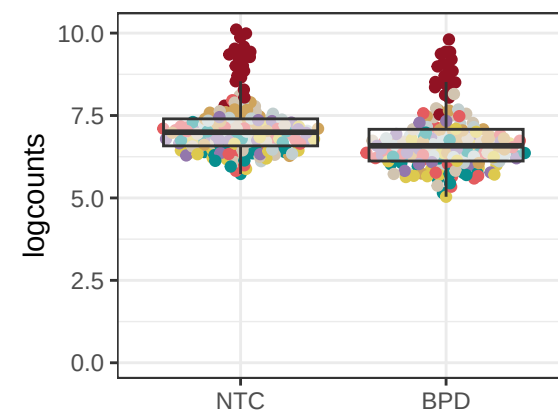
PRECAST (smoothed)

L-A sig: adj p < $1\text{e-}04$ logFC = -0.41



Seurat labels

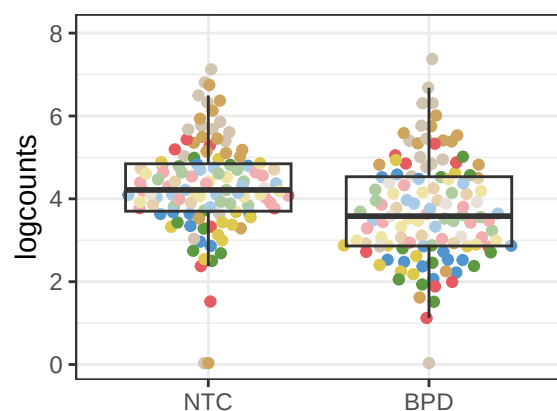
L-A sig: adj p < $7\text{e-}06$ logFC = -0.41



HLA-DPA1

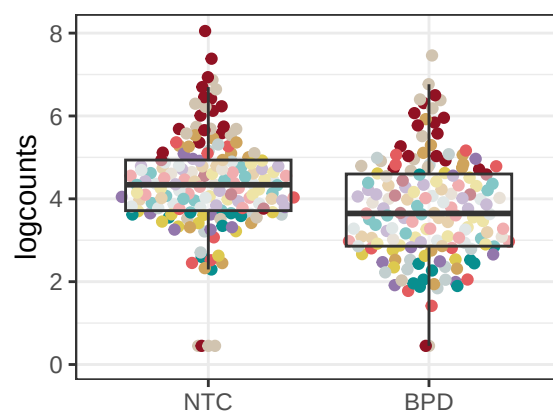
PRECAST (smoothed)

L-A sig: adj p < $6\text{e-}04$ logFC = -0.76



Seurat labels

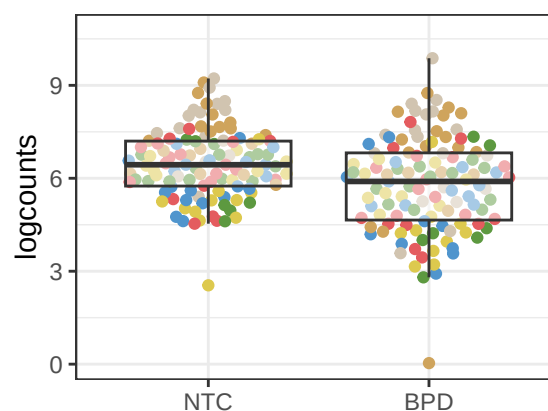
L-A sig: adj p < $2\text{e-}04$ logFC = -0.73



CD74

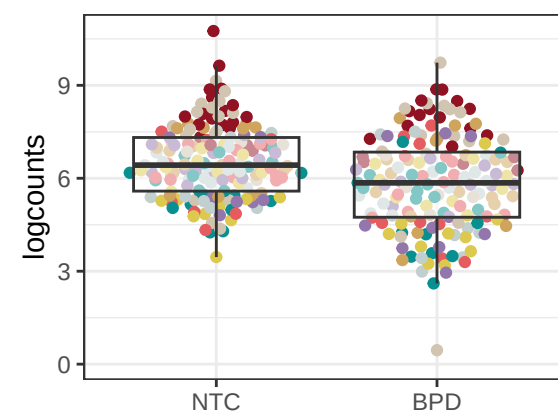
PRECAST (smoothed)

L-A sig: adj p < $4\text{e-}03$ logFC = -0.65



Seurat labels

L-A sig: adj p < $5\text{e-}04$ logFC = -0.64

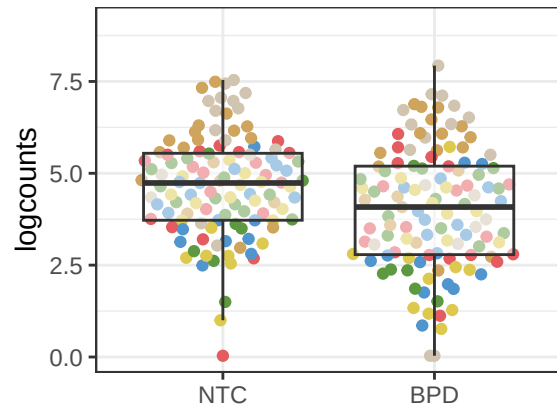


All genes with $\text{abs}(\log\text{FC}) > .5$

HLA-DRA

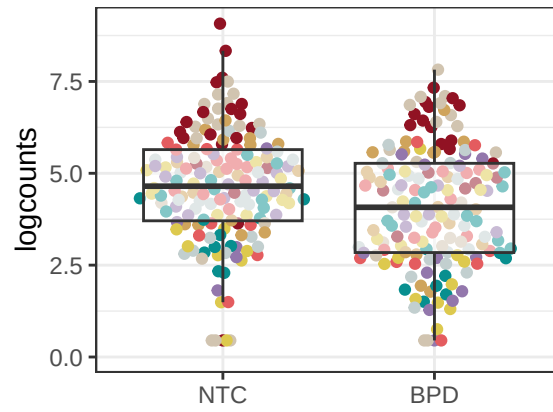
PRECAST (smoothed)

L-A sig: adj $p < 1\text{e-}02$ $\log\text{FC} = -0.72$



Seurat labels

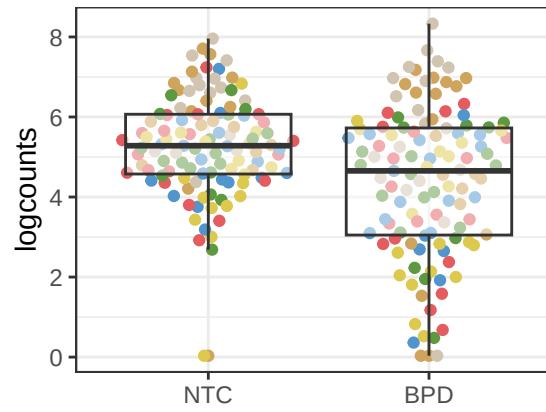
L-A sig: adj $p < 4\text{e-}03$ $\log\text{FC} = -0.74$



C3

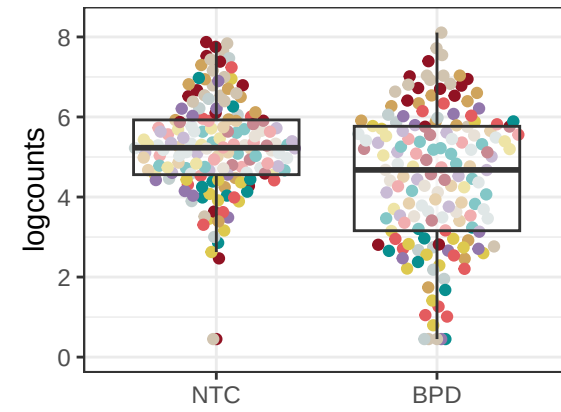
PRECAST (smoothed)

L-A sig: adj $p < 3\text{e-}04$ $\log\text{FC} = -0.97$



Seurat labels

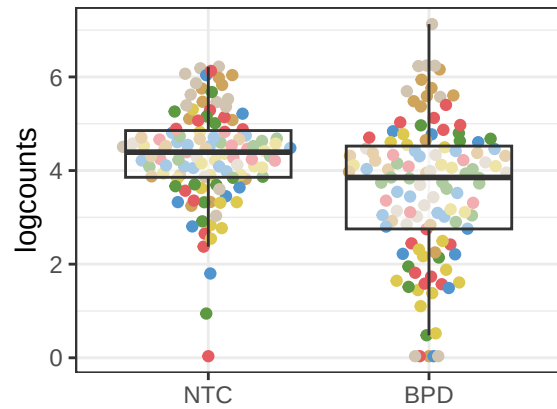
L-A sig: adj $p < 2\text{e-}05$ $\log\text{FC} = -1$



CSF1R

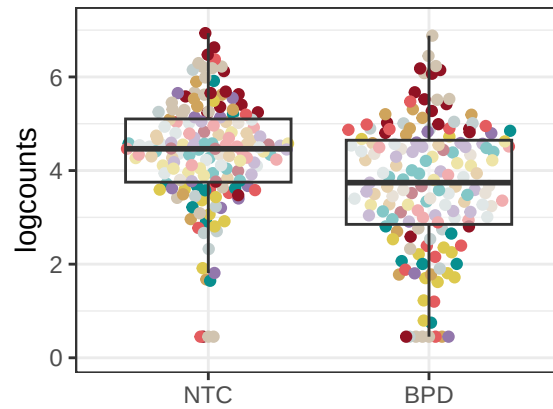
PRECAST (smoothed)

L-A sig: adj $p < 1\text{e-}04$ $\log\text{FC} = -0.93$



Seurat labels

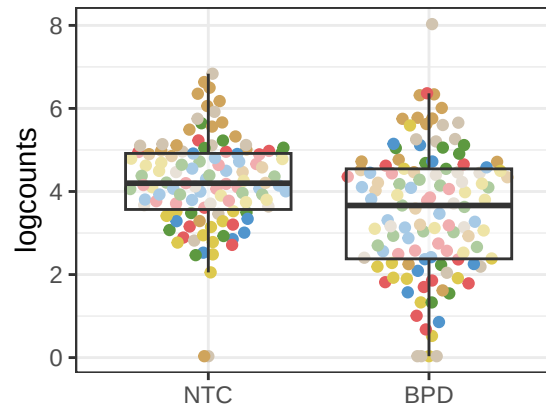
L-A sig: adj $p < 3\text{e-}05$ $\log\text{FC} = -0.9$



C1QA

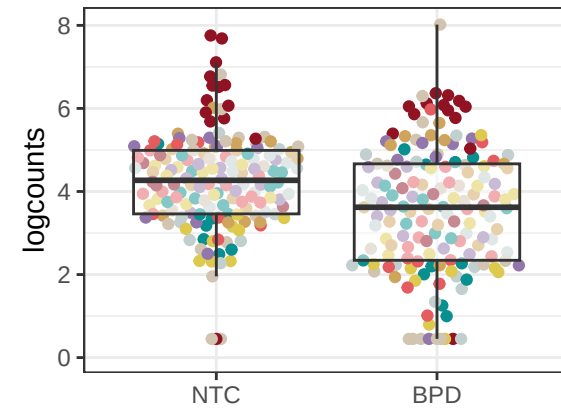
PRECAST (smoothed)

L-A sig: adj $p < 1\text{e-}02$ $\log\text{FC} = -0.74$



Seurat labels

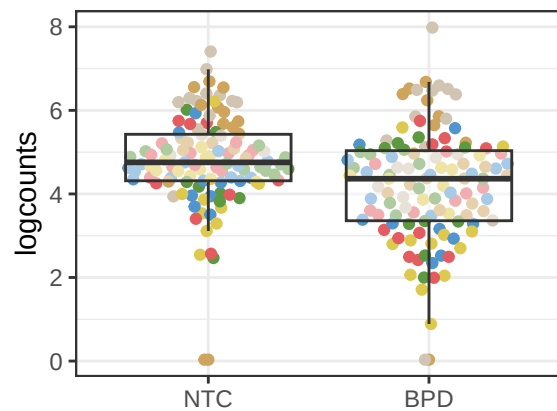
L-A sig: adj $p < 4\text{e-}03$ $\log\text{FC} = -0.74$



LAPTM5

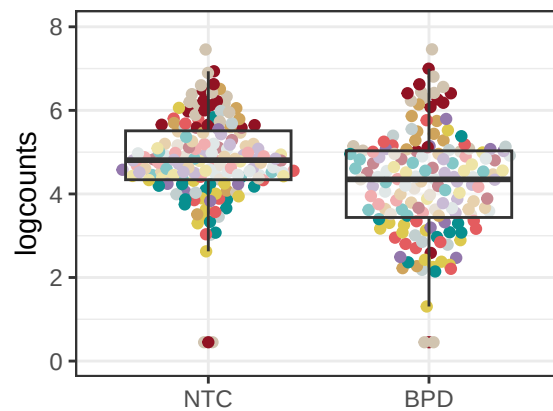
PRECAST (smoothed)

L-A sig: adj $p < 4\text{e-}03$ $\log\text{FC} = -0.64$



Seurat labels

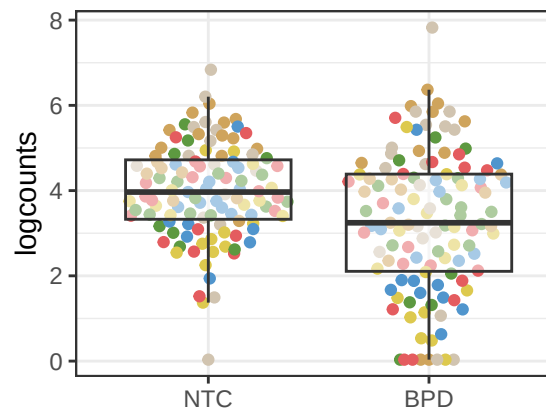
L-A sig: adj $p < 7\text{e-}04$ $\log\text{FC} = -0.66$



AIF1

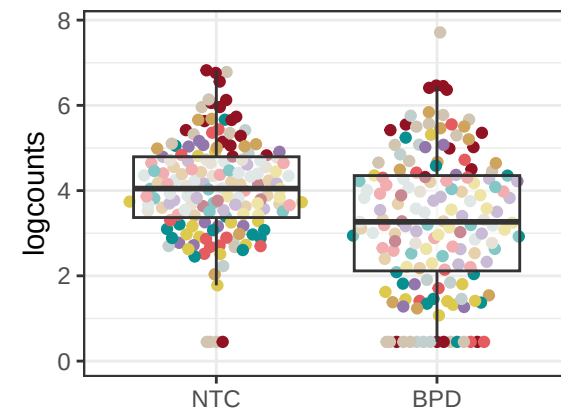
PRECAST (smoothed)

L-A sig: adj $p < 2\text{e-}02$ $\log\text{FC} = -0.75$



Seurat labels

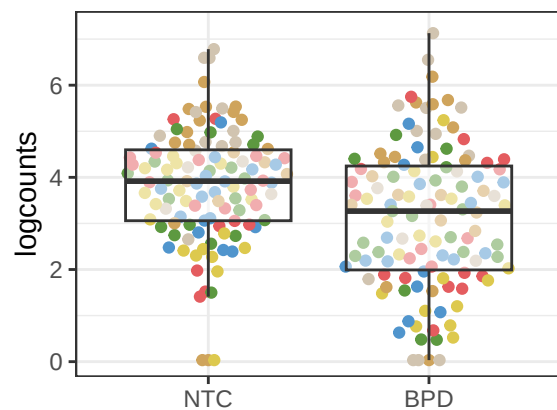
L-A sig: adj $p < 2\text{e-}02$ $\log\text{FC} = -0.69$



TYROBP

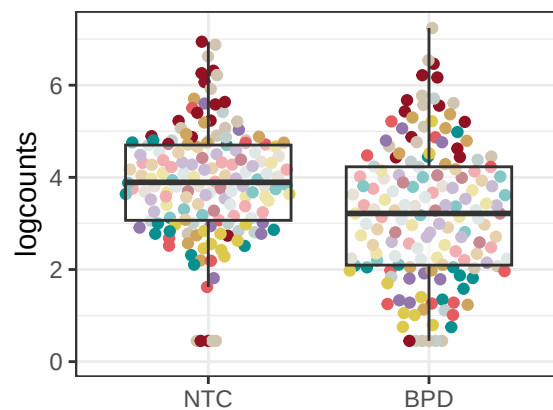
PRECAST (smoothed)

L-A sig: adj $p < 1\text{e-}02$ $\log\text{FC} = -0.73$



Seurat labels

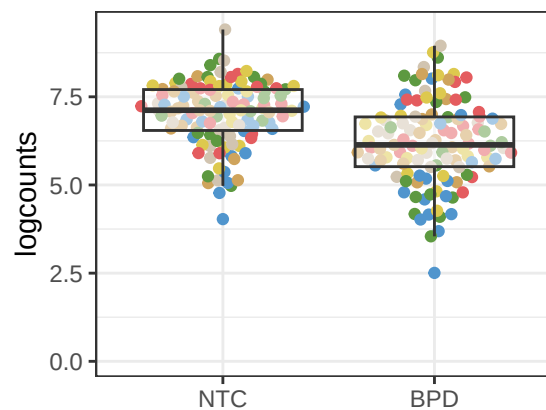
L-A sig: adj $p < 2\text{e-}03$ $\log\text{FC} = -0.79$



SST

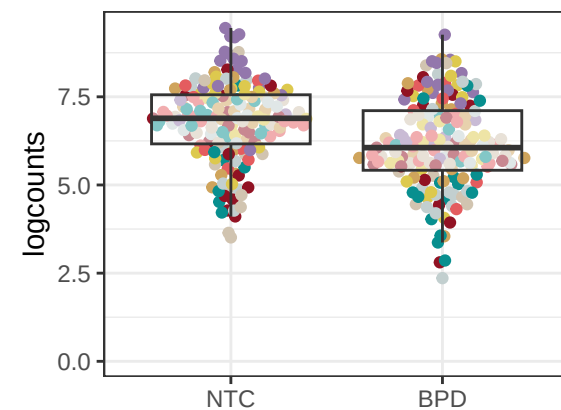
PRECAST (smoothed)

L-A sig: adj $p < 2\text{e-}03$ $\log\text{FC} = -0.81$



Seurat labels

L-A sig: adj $p < 1\text{e-}03$ $\log\text{FC} = -0.75$

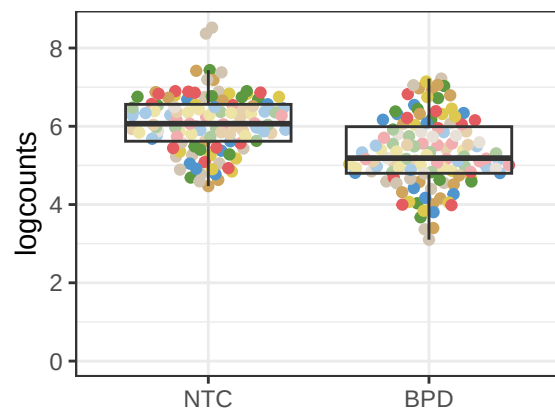


All genes with $\text{abs}(\log\text{FC}) > .5$

CORT

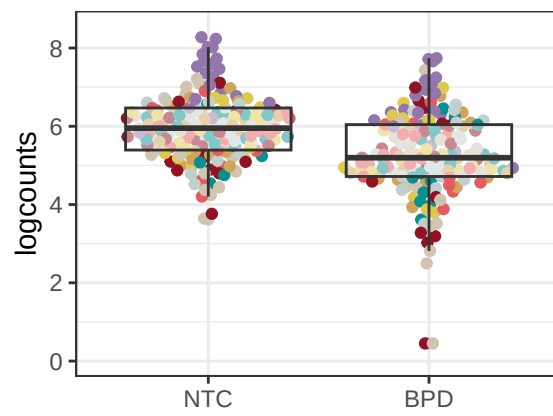
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}04$ logFC = -0.61



Seurat labels

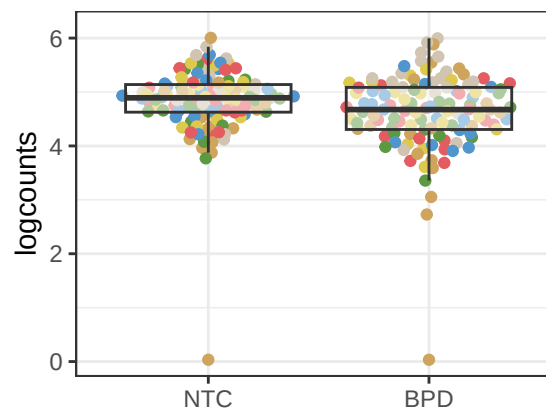
L-A sig: adj p < $2\text{e-}04$ logFC = -0.57



PIGP

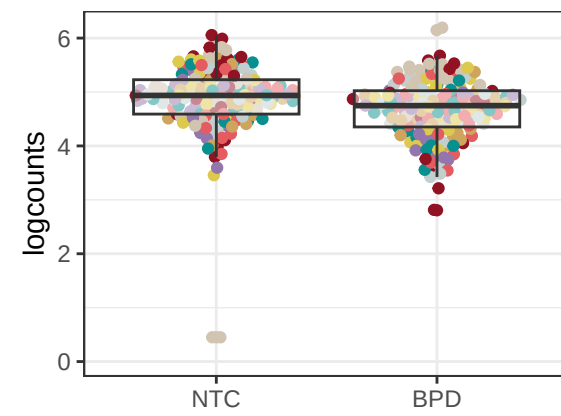
PRECAST (smoothed)

L-A sig: adj p < $2\text{e-}02$ logFC = -0.31



Seurat labels

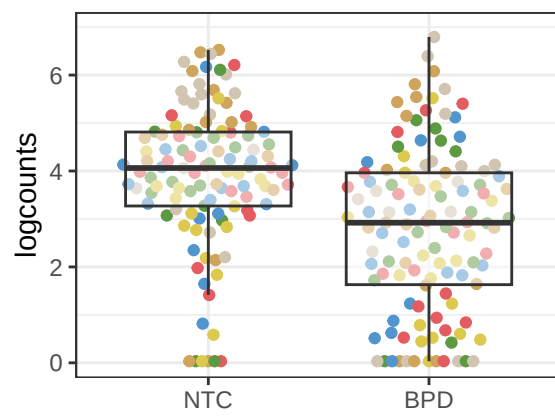
L-A sig: adj p < $3\text{e-}02$ logFC = -0.32



CX3CR1

PRECAST (smoothed)

L-A sig: adj p < $1\text{e-}07$ logFC = -1.52



Seurat labels

L-A sig: adj p < $2\text{e-}08$ logFC = -1.47

